

Mars habitat: A holistic approach to design extraterrestrial structures under extreme environments

Pietro Lapo Moro^{a, *}, Klajdi Toska^{a, *}, Mariano Angelo Zanini^a, Maurizio Pajola^b, Flora Faleschini^{a, *}

^a Dept. of Civil, Environmental and Architectural Engineering, University of Padova, Via Marzolo 9, Padova, 35131, Italy

^b INAF, Astronomical Observatory of Padova, Vic. Osservatorio 5, Padova, 35122, Italy

ARTICLE INFO

Keywords:

Asteroid impacts
Extraterrestrial habitat
Mars
Marsquake
Radiation
3D printing

ABSTRACT

Martian exploration represents one of the most ambitious goals of the 21st century. What only a few decades ago might have seemed like a futuristic vision has now become a tangible objective, thanks to rapid technological development and the growing interest of scientific programs. This paper fosters the ambitious goal of developing an interdisciplinary study that brings together engineering, material and geology sciences, as well as planetology, to propose a new Martian habitat designed through a rational and scientific approach. A new framework for extraterrestrial habitats design is thus conceived and applied to the case of Mars. Once defined the mission type and the landing site, the main environmental aspects that impact on the structural and architectural design of a structure on Mars are identified. Among them, reduced gravity, seismic activity, radiation, wind and asteroids' impacts need to be considered. Then, a review on the locally-available construction materials is carried out. This is done by exploring also the construction methods supported by the most recent technological advancements carried by additive manufacturing in extreme environments. Lastly, a proposal for a new habitat module is made, which is constituted by an external 3D printed shell, whose structure has been verified according to the proposed design scenario through finite element (FE) analyses, and inner inflatable-balloon modules. Among the significant outcomes of this research, a special mention has to be made to the in-depth analysis of design aspects on Mars, thus representing fundamental inputs for the structural analysis of the new habitat.

1. Introduction

Mars has long represented a key focus for scientific investigation, whose research has contributed significantly to the development of astronomy and planetary sciences. Today, international missions such as Mars Express [1], Mars Reconnaissance Orbiter [2], ExoMars Trace Gas Orbiter [3] and MAVEN [4] and technological progress sustain growing interdisciplinary research on the planet's present and past geological evolution, atmosphere, and potential habitability. The body of knowledge acquired so far both from orbit and from in situ analyses [5] has confirmed the past presence of liquid water, current polar ice deposits, and the absence of a global magnetic field (only of the presence of numerous local magnetic fields is nowadays confirmed [6]), which led to atmospheric erosion by the solar wind. At the same time, in-situ resource utilization (ISRU) experiments have demonstrated the feasibility of producing oxygen from atmospheric CO₂, opening perspectives for future human exploration. Building upon this scientific evidence, a

growing need is emerging, dealing with the necessity to develop a methodological approach for designing reliable habitats that can sustain loads and actions present on Mars. In this context, the use of in-situ materials, such as Martian regolith-based composites, is particularly attractive, as it reduces the mass and cost of transported materials, takes advantage of locally available resources, and enables the construction of structurally efficient elements tailored to the Martian environment.

Among the initiatives aimed at demonstrating the feasibility of extraterrestrial construction, the NASA 3D-Printed Habitat Challenge (NMHC), launched in 2015, represents an important effort to promote additive manufacturing systems capable of using in-situ resources to build habitat structures. The competition engaged teams in architectural conceptualization, material formulation, and the development of printing strategies for scaled prototypes [7–9]. However, the NMHC primarily focused on the technological demonstration of 3D printing feasibility rather than a comprehensive structural design process that integrates environmental loads, material behavior, and full verification

* Corresponding author.

E-mail address: flora.faleschini@unipd.it (F. Faleschini).

<https://doi.org/10.1016/j.actaastro.2026.06.008>

Received 30 December 2025; Received in revised form 8 May 2026; Accepted 3 June 2026

Available online 4 June 2026

0094-5765/© 2026 The Authors. Published by Elsevier Ltd on behalf of IAA. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

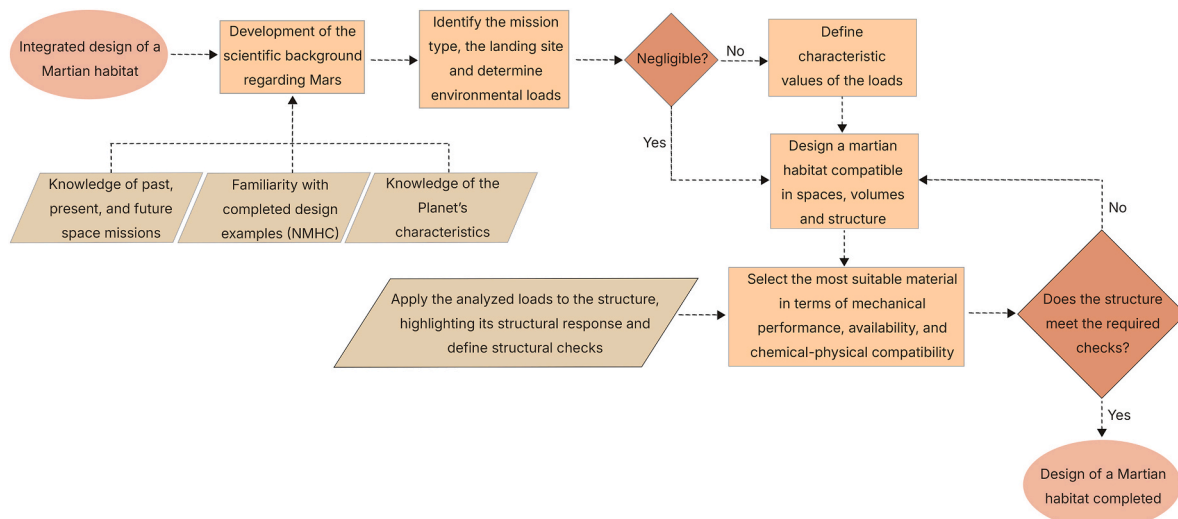


Fig. 1. Flowchart of the proposed methodological approach.

procedures. Recent research has begun addressing structural and performance aspects of Martian habitat design. Soureshjani and Massumi [10] investigated structural forms optimized for Martian conditions using in-place binders such as sulfur-based concrete to minimize tensile stresses under Martian loads, including gravity, internal pressure, wind, marsquakes, and meteoroid impacts, and compared these with conventional concrete forms to improve structural resilience and reduce importation of Earth-based elements. Zhou et al. [11] introduced the Zhuque Base concept, drawing on cave-dwelling principles and finite-element analysis to evaluate multiple arch-based configurations for mechanical robustness and thermal efficiency in Martian environments, thereby highlighting trade-offs between structural performance and habitat energy behavior. Koca and Turer [12] proposed an approach to design, optimize, and autonomously construct Martian infrastructure using in-situ CO₂-derived polyethylene materials, explicitly considering environmental challenges such as low pressure, radiation, micrometeoroid impacts, and marsquake effects, and demonstrating how novel material systems can influence design optimization and energy considerations. Despite these contributions, current literature remains limited in proposing a unified, systematic methodological framework that integrates environmental characterization, mission definition, architectural configuration, material selection, and structural verification within a single workflow tailored to Martian habitat design. This gap motivates the comprehensive framework proposed in this study, thus leading to a design framework for a Martian habitat that covers all pivotal stages from the analysis of planetary conditions to complete structural verification.

1.1. Methodological framework

The structure of this framework is not arbitrary but derives from the strong interdependence between environmental constraints, mission requirements, architectural configuration, and structural performance in extraterrestrial contexts. The methodological approach was indeed developed and subsequently refined through an in-depth review of the relevant scientific literature, which is referenced progressively throughout the manuscript. This body of knowledge has been adapted to account for the complexity and full set of constraints associated with the Martian environment.

Infact, unlike terrestrial design processes -where established regulations provide predefined load prescriptions and material standards- Martian habitat design requires a bottom-up approach in which planetary characterization precedes architectural and structural decisions. For this reason, the workflow is organized in a sequential yet iterative

manner, ensuring that environmental and mission constraints govern subsequent design choices. The overall workflow is summarized in the flowchart shown in Fig. 1. As previously stated, the methodology begins with the development of a solid scientific background on the Martian environment, including knowledge from space missions, familiarity with completed habitat design examples, and an understanding of the planet's physical characteristics, in order to identify relevant external constraints and actions. The mission type and landing site are then defined, as they determine the project's strategic direction and the site-specific environmental phenomena acting over the structure's lifetime. Based on these inputs, architectural forms, volumes, and spatial configurations are explored to define the habitat concept. Finally, material selection -considering mechanical performance, potential in-situ availability, and environmental compatibility- is carried out alongside structural verification. The analyzed loads are applied to assess structural response and perform required checks, with an iterative loop allowing refinement of design choices until performance requirements are satisfied.

2. Background on Mars exploration

A comprehensive understanding of the Martian planetary context represents a fundamental prerequisite for the development of a coherent architectural and structural design process. The planet's physical characteristics, environmental conditions, and the scientific evidence collected through past and ongoing exploration missions provide the essential knowledge base for the correct identification of constraints, design actions, and construction strategies.

This section therefore summarizes the main planetary data and the most relevant outcomes of Mars exploration, with the aim of outlining the technical and scientific framework within which the proposed methodological approach is developed. These elements constitute the reference foundation for the subsequent analyses concerning load definition, material selection, and habitat configuration.

2.1. Mars exploration

Over the years, space agencies have focused their programs on numerous missions aimed at exploring Mars and gathering valuable information about its geology, atmosphere, surface and subsurface [13, 14]. However, roughly half of these missions have ended in failure due to technical issues or technological malfunctions. Here we mention the most important past, present, and future missions undertaken to study the Red Planet. Beginning with NASA's Mariner 4 in 1964 [15], which

Table 1
Comparison of the main physical characteristics of Earth and Mars.

Parameter	Earth	Mars
Surface area (m ²)	5.1·10 ¹⁴	1.45·10 ¹⁴
Mean density (kg/m ³)	5.5·10 ³	3.9·10 ³
Surface gravity (m/s ²)	9.78	3.73
Albedo	0.37	0.15
Maximum surface temperature (K)	331	293
Minimum surface temperature (K)	185	133
Atmospheric pressure (Pa)	101300	8000

captured the first close-up images of the Martian surface, both the United States and the Soviet Union advanced their efforts throughout the 1970s with mixed success. Notably, the Viking missions (1975) [16] provided the first color images and surface maps. Subsequent probes such as the Soviet Phobos (1988) [17], NASA's Mars Global Surveyor (1996) [18], and Mars Odyssey (2001) [19] contributed with crucial data on the planet's topography and hydrogen-rich regolith. In 2003, the European Space Agency's (ESA) Mars Express Mission [1], equipped with radar and spectrometers, revealed polar water ice and possible subsurface brines, while NASA's Mars Reconnaissance Orbiter (2005) [2] delivered high-resolution (meter to centimeter spatial scale) surface imaging. It is also pivotal to mention the NASA missions that have revolutionized the scientific world: the twin rovers Spirit and Opportunity (2004) [20] and the Phoenix Lander (2007) [21] which confirmed the presence of water ice. Such rovers and lander paved the way for Curiosity (2012) [22], which detected complex chemistry in Gale Crater. ESA's ExoMars program (2016) [3] developed and sent the Trace Gas Orbiter for atmospheric studies, though its lander Schiaparelli failed upon descent. More recently, the global scope of Mars exploration expanded with China's Tianwen-1 mission (2020) [23], comprising an orbiter, lander, and rover (Zhurong), and the United Arab Emirates' Hope Probe (2020) [24], dedicated to studying the planet's climate and atmospheric dynamics. Together, these missions have transformed Mars from a distant target of curiosity into an increasingly well-characterized world of scientific inquiry [25].

2.2. Planetary data

Mars is the fourth planet of the Solar System, characterized by extreme environmental conditions that are critical for habitat design. Surface gravity is 3.73 m/s², and temperatures range from −140 °C to +20 °C (average −63 °C) [26], influenced by a thin atmosphere composed mainly of carbon dioxide (95%) at 6.1 mbar, producing only a

weak greenhouse effect [27]. The crust beneath the InSight landing site consists of at least two, possibly three layers, with local thickness ranging from 20 to 39 km, while global estimates place the average Martian crust thickness between 24 and 72 km [28]. Mars lacks a global magnetic field due to its inactive core, which, together with low atmospheric pressure, exposes the surface to high levels of cosmic and solar radiation. The planet's surface is also constantly reshaped by dust storms driven by atmospheric dynamics, which can affect surface operations and material erosion [29]. Table 1 summarizes the main environmental parameters of Mars that are directly relevant for habitat design, providing a concise reference for structural, thermal, and shielding considerations.

3. Mars habitat design

3.1. Mission design

The first step to design a habitat requires choosing two key elements: the entire mission duration [30] and the landing site [31–33]. Regarding the first aspect, a mission must balance multiple objectives: minimizing risks to the crew and costs, ensuring sufficient time for scientific activities, and optimizing return. Two classes of Mars missions with return to Earth are distinguished [30]: Opposition Class missions, characterized by short surface stays, and Conjunction Class missions, with longer durations, also known as long-stay missions. Opposition Class missions have surface stays of 30–90 days and relatively short transit durations of 400–600 days, defined as short-stay (Fig. 2). These missions require higher propulsive capabilities and often utilize gravitational fly-bys of Venus or complex deep-space maneuvers. Conjunction Class missions, on the other hand, have a total duration of approximately 900 days and represent minimum-energy solutions for the given launch window, with extended surface operations on Mars (Fig. 2). For short-stay missions, over 90% of the total mission time is spent in transit, compared to only about 30% for long-stay missions, highlighting that rapid transit windows are primarily available for long-duration missions. A critical factor for the success of an interplanetary mission is the crew's exposure to radiation. Both mission types present radiation levels significantly above the acceptable limits for human health. It is estimated [30] that for short-stay missions, the estimated risk of death due to radiation exposure (REID) is 7.8%, while for long-stay missions, this risk is approximately 8%. In optimal conditions, i.e. for Galactic Cosmic Radiations (GCR) at solar maximum, the total Blood-Forming Organ (BFO) exposure ranges between 450 and 460 or 395–450 mSv for short- and long-stay missions, respectively. These values represent the mean of

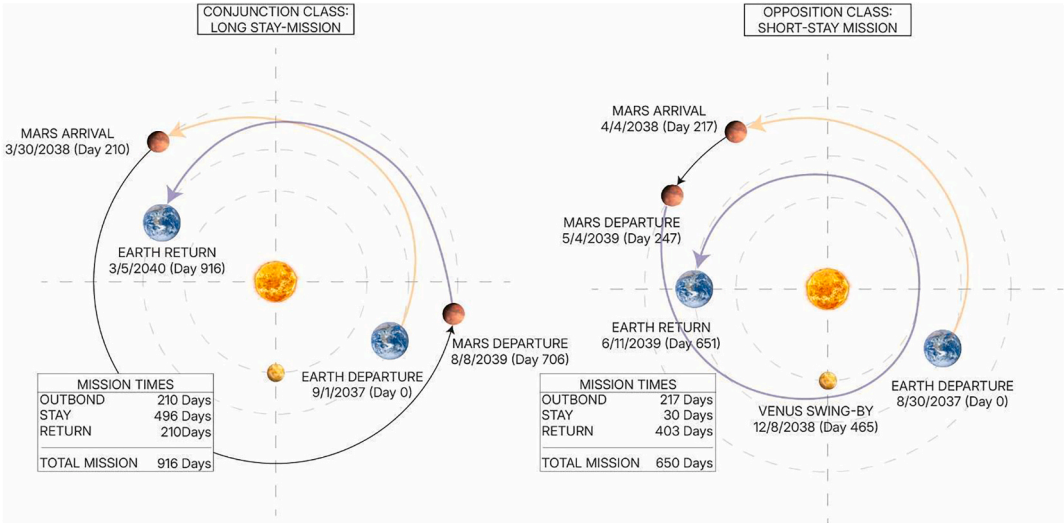


Fig. 2. Example of the timeline of a long-stay mission and of the timeline of a short-stay mission. Adapted from Drake et al. [30].

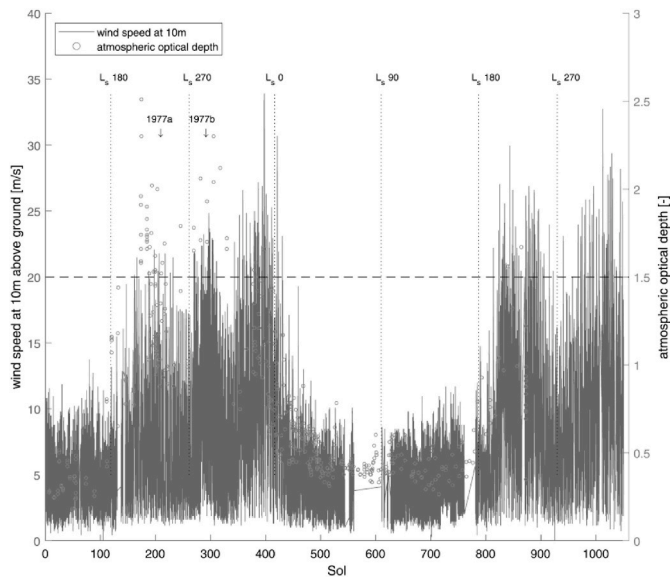


Fig. 3. Wind speed at an altitude of 10m and atmospheric optical depth measured, both plotted over the entire duration of the Viking Lander 2 mission. Adapted from: Schorbach & Weiland [35].

the results within a 95% confidence interval, meaning there is only a 5% probability that the actual values are higher. However, considering the upper bound of the interval, both mission types would face a radiation-induced mortality risk of 16% REID. Given that NASA sets a maximum acceptable REID of 3% at the 0.95 percentile, mitigation strategies are required. Additional factors that increase risk include:

- Non-targeted effects in cancer induction during both short- and long-duration missions;
- Limited knowledge regarding the protective effect of the Martian surface and atmosphere, particularly relevant for long-duration missions;

- The potential for exposure to Solar Particle Events (SPEs) during maneuvers near the Sun, especially in short-stay missions utilizing Venus swing-by;
- The still poorly understood risk of damage to the central nervous system and degenerative tissues.

Instead, regarding the second aspect, selecting a landing site on Mars is another critical phase, requiring the reconciliation of engineering constraints and scientific objectives [34]. Engineering requirements concern all aspects related to safety and technical feasibility, thus they include factors related to the safety of launch, transit, and landing, such as latitude, elevation, landing ellipse, terrain slope, rock abundance, and the thermal and atmospheric features of the site [31]. Scientific requirements, on the other hand, aim to identify areas of interest, such as locations that may preserve traces of past life [31]. Therefore, the final site selection represents a compromise between safety and scientific objectives, and the following section analyzes the main engineering factors that determine this choice [34].

3.2. Loads definition

Structural design on Earth is based on the analysis of the loads acting at the project site, including the evaluation of the effects of atmospheric phenomena such as wind and snow, dead and live loads related to the building's function, and the effect of further actions such as accidental loads (blast, fire) or earthquake occurrence. On Mars, the effects of these actions may differ significantly, thus their relevance to the structural design should be first evaluated. Furthermore, other environmental effects may be relevant, such as the case of radiation and the potential impacts from micrometeorites and asteroids, both related to the peculiarity of Mars atmosphere. Moreover, another important environmental load relates to the formation of dust storms.

In this context, we have evaluated that the following aspects may have a relevant impact on the structural design of a Mars habitat: dead load, Martian wind, asteroid impacts, radiation, marsquake activity and thermal stresses due to high temperature fluctuations. Regarding the dead load, reduced gravity conditions should be applied for the proper estimation of this action.

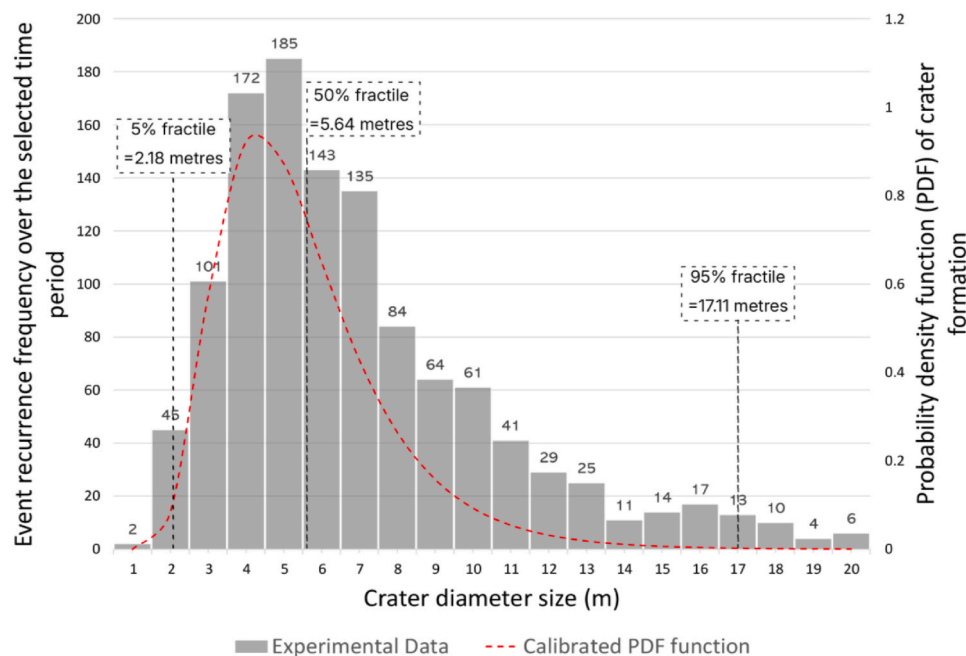


Fig. 4. Probability density function of the crater diameters, calibrated on the experimental data collected from Daubar et al. [39]. The 5, 50 and 95% fractiles are highlighted.

3.2.1. Wind load

On Earth, wind represents a major load in structural design that can be treated both as equivalent-static or dynamic action. On Mars, however, wind plays a substantially different role. Average wind speeds on the Martian surface are slightly below 20 m/s, with extreme cases reaching 35–37 m/s [35,36], as illustrated in Fig. 3. Local dust storms can reach up to approximately 97 km/h [35,36], less than one-third of the velocity observed in the most intense terrestrial hurricanes [37]. However, to translate wind speed effects onto an applied load to a structure and calculate wind-induced pressure, it is worth recalling that atmospheric density plays a key role. The Martian atmosphere is extremely thin, composed predominantly of CO₂, with a surface pressure less than 1% of Earth's. Even assuming a wind speed of 26.8 m/s, the resulting dynamic pressure is negligible for non-inflatable structural systems. In fact, assuming an area of 1 m² and an atmospheric density of 0.020 kg/m³, the resulting pressure is 7.4·10⁻¹⁰ MPa, which is completely negligible. Nevertheless, dust storms may damage in-situ machinery and operational systems by covering solar panels with fine dust, reducing energy efficiency, and limiting rover operations, as documented in the cases of Spirit and Opportunity [38]. Consequently, wind loads are neglected within the structural design due to their minimal impact. However, future studies should address mitigation strategies, such as designing automatic cleaning mechanisms for solar panels, to ensure the reliability of on-site equipment and structures.

3.2.2. Potential impacts from micrometeorites and asteroids

The thin Martian atmosphere provides far less protection against asteroid impacts than Earth's. To assess the probability of collapse of a hypothetical habitat due to an impact, first of all a revision about impacts' records is carried out. Afterwards, their consequences on a hypothetical structure are considered.

The occurrence of an impact event on Mars is detected exclusively a posteriori, and it is based on a visual analysis of the surface in terms of presence and quantification of craters' dimension. This approach stems from the limited direct data on asteroid mass, diameter, and density, which are instead estimated *ex post* through crater analysis. Even the duration of the time series of observations is limited. Among the few available catalogs, Daubar et al. [39] report over 1200 new craters formed on Mars between 7 March 2008 and 11 February 2021. Craters were classified in 1-m diameter intervals and analyzed according to detection frequency. From the observed data, shown in the histogram of Fig. 4, a lognormal probability distribution was fitted (red color, in the same Figure). The 5%, 50% and 95% fractiles of the distribution are, respectively, 2.18, 5.64 and 17.11 m, which indicate that most of the craters have a dimension higher than 3 m. Furthermore, the distribution has a slight right-skewed tail due to the lower probability of extreme events. Considering the limited time window of 13 years of observation, it is however recognized that this tail may be underestimating crater dimensions in case of great asteroids' impact. Thus, the consequence of an impact on a habitat may have devastating effects on the habitat, and a proper evaluation regarding the assessment of potential occurrence of such an event should be carried out.

3.2.3. Radiation

Any extraterrestrial habitats should guarantee the security and safety of its inhabitants, in this case the main risk comes from the hazard of radiation. Indeed, radiation exposure poses both long-term risks to astronauts, such as cancer, cognitive and motor function impairments, and neurological disorders, and short-term mission-related risks, including nausea, fatigue, and loss of consciousness [30]. The type and source of radiation may come from Refs. [30,40]:

- Galactic Cosmic Rays: fully ionized particles originating from outside the Solar System, mainly composed of protons (91%), alpha particles (8%), and heavy charged particles (1%);

- Solar Particle Events: temporary and sudden ejections of particles, primarily protons, associated with solar flares or Coronal Mass Ejections (CME), exhibiting high particle fluxes but lower kinetic energies than GCRs. CME frequency varies with the solar cycle, reaching up to three events per day at solar maximum and averaging one every five days at solar minimum;
- Trapped Radiation Belts (Van Allen Belts): cosmic radiation particles trapped along Earth's magnetic field lines, primarily protons and electrons;
- Secondary Planetary Radiation: generated by the interaction of radiation spectra with planetary atmospheres or surfaces, predominantly neutrons.

Among the possible types of radiation sources, GCR and SPE are predominant: the former include nuclei of nearly all known chemical elements and may reach energies up to 10¹² MeV per nucleon; while the latter, despite lower energies, involve significantly higher particle fluxes. Thus, the habitat requires a proper design of a specific shielding layer to lower radiation's effect for ensuring astronauts' security.

3.2.4. Seismic load

Martian crust, while showing many similarities to Earth's, is currently tectonically inactive [41]. Nevertheless, certain linear magnetic anomalies and fault zones, such as the large Valles Marineris depression, suggest that some regions may have experienced limited tectonic activity in the past [42]. The study of Yin et al. [42] compares these structures to terrestrial mid-ocean ridges or transform faults like the San Andreas and Dead Sea faults, indicating potential localized crustal movements in ancient geological epochs. However, current seismic activity on the planet appears more related to anisotropic cooling of Mars' interior rather than to tectonic movements, generating differential stresses on the crust [42]. Independently from the source of such quakes, seismic data remain limited, because Mars lacks a wide-spread seismometer network covering all the planet. Only recently, seismic activity on Mars has been systematically recorded, starting from the landing of the InSight mission on 26 November 2018. One of the most complete datasets available is the one provided by IRIS DMC [43], that compiles all seismic events recorded on Mars. For each event, it provides the date and time of onset, together with the event's designation and classification (Low Frequency, BroadBand, 2.4 Hz, High Frequency, Very High Frequency). When available, the catalog also includes additional parameters such as epicentral distance (expressed in degrees), magnitude, data quality rating ranging from A to D (letters state for: A - high quality, with a clear signal for which both distance and back-azimuth can be reliably determined; B - medium quality, with a clear signal and a reliable distance estimate but poorly constrained direction; C - low quality, with only a fair signal making both distance and direction difficult to determine; and D - indeterminate, with a weak signal for which neither distance nor direction can be reliably inferred), and the arrival times of the main seismic phases, including both primary and secondary waves. Moreover, the mentioned dataset provides access to downloadable GeoCSV and miniSEED files containing experimental instrument data. These data enable users to compute the parameters required for deriving Peak Ground Velocity (PGV) and Peak Ground Acceleration (PGA).

Despite the availability of the above-mentioned data, the localization of marsquakes is often subject to some uncertainty, as single seismometers cannot provide precise triangulation, and are used just for identifying regions relatively more active. Among them, worth mentioning is the Cerberus Fossae, which is one of the most seismically active regions on Mars, with over half of the analyzed low-frequency events located in this area [44]. In particular, Horvath et al. [45] and Taylor et al. [46] indicate that this region is a geologically young (~20 million years) tectonic trench likely formed by magmatic intrusion [47,48], characterized by a weak structure and localized heat anomalies [49]. However, even outside this region the evidence of the occurrence of relevant

Table 2

Temperature trends during the two solstices and the two equinoxes at the Vernal Crater site, based on data provided by Pajola et al. [50].

Time	Ls = 0° (Northern hemisphere spring equinox) [K]	Ls = 90° (Summer Solstice) [K]	Ls = 180° (Autumn Equinox) [K]	Ls = 270° (Winter Solstice) [K]
0	170	175	158	173
2	165	171	157	170
4	163	168	155	167
6	160	165	150	165
8	233	223	245	215
10	274	259	282	264
12	287	272	292	280
14	278	268	281	271
16	250	244	246	240
18	200	205	182	196
20	181	187	168	182
22	174	180	162	176
24	170	174	158	173

quakes requires a proper seismic design of the habitat.

3.2.5. Thermal load

The influence of thermal loads on the structural analysis of the habitat was also taken into account. Seasonal temperature variations on the Martian surface are, in fact, substantial, as confirmed by the data reported in the study by Pajola et al. and by Munaretto et al. [50,51] concerning the temperatures recorded at the surface of Vernal Crater. The study examined not only significant fluctuations across the seasons, but, more importantly, a marked variability in the diurnal temperature oscillations observed. As shown in Table 2, the largest temperature variations occur around midday, particularly during the summer solstice ($L_s = 90^\circ$), reaching peaks as high as 305 K, while the coldest temperatures are observed at dawn during the autumn equinox ($L_s = 180^\circ$), dropping to 150 K. These fluctuations are expected to induce thermal stresses and strains in the habitat structure, which must be accounted for in order to prevent material fatigue or structural failure. Furthermore, these fluctuations could impact sensitive onboard equipment or instruments, requiring careful placement and thermal shielding strategies.

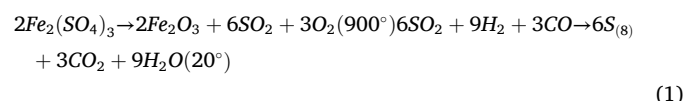
3.3. Material and construction technique selection

Material selection is always a fundamental step in the design process, as mechanical and durability-related properties impact directly on final structural behavior. On Mars, this choice becomes even more important, as it should be guided not only on targeted material properties, but even by considerations linked to materials' availability and ease of transportation and manufacture. Indeed, one of the main problems that may be faced during the realization of an extraterrestrial habitat relates to resource scarcity. For example, to use traditional construction material, such as concrete, the non-availability of liquid water required for cement hydration and fresh concrete pumping and casting becomes a key challenge that makes almost impossible to adopt this choice. The same consideration applies for the selection of the best construction

process: if we rely on traditional construction methods, it becomes clear that the realization of the habitat becomes almost impossible. Based on a wide literature review, it was found that many so-called “marscretes” were proposed, and most of the studies affirm the suitability of adopting sulfurcrete as a valid construction material on Mars [52–55]. The construction methodology, instead, will make use of additional manufacturing processes, as investigated in the recent NASA Mars Challenges [7].

Sulfur concrete has been investigated since the 1920s, demonstrating high compressive strength and acid resistance [56]. When molten sulfur is combined with aggregate and subsequently cooled, the binder crystallizes first as monoclinic sulfur (Sb) at 114°C (238°F) and then transforms into stable orthorhombic sulfur (Sa) below 96°C (204°F), completing hardening within 24 h [52]. Compared to conventional concrete, which requires approximately 28 days to achieve the targeted strength, sulfur concrete provides significant advantages: improved mechanical performance, chemical and freeze–thaw resistance, rapid setting, greater tolerance in aggregate selection, reduced environmental impact (no water consumption, low CO_2 footprint), and economic benefits. In 1993, experiments using lunar regolith simulants mixed with varying sulfur contents (25–75%) showed that the optimal composition was 35% sulfur, achieving a compressive strength of 34 MPa. The addition of 2% steel fibers further increased strength to 43 MPa [57]. Nonetheless, the lunar environment - with its near-vacuum conditions, wide thermal variations, and long day–night cycles - proved unsuitable for large-scale applications. Conversely, Mars offers a more favorable scenario. In particular, Mars contains significant sulfur deposits, likely in the form of sulfide minerals and sulfate salts, which can be extracted through physicochemical processes. With an atmospheric pressure of 0.636 kPa and more moderate temperature ranges, sulfurcrete could be employed without the risk of binder melting at temperatures below 35°C .

Specifically dealing with a hypothetical sulfurcrete production process, water would be replaced by pure sulfur, which can be sourced from sulfate minerals widely available on the planet's surface, such as ferric sulfate ($\text{Fe}_2(\text{SO}_4)_3$) [58]. The extraction process involves the pyrolysis of ferric sulfate followed by the catalytic reduction of sulfur dioxide (SO_2). The synthesis of 1 kg of elemental sulfur requires approximately 22.5 g of H_2 and 105 g of CO , both of which can be derived from in-situ water and atmospheric CO_2 . The reaction (Eq. (1)) products include 240 g of sulfur, 203 g of water, and 120 g of oxygen, with by-products consisting of 400 g of Fe_2O_3 and 165 g of CO_2 . Notably, both water and CO_2 may be recycled to regenerate H_2 and CO , thereby enabling a quasi-closed production loop that minimizes resource consumption.



Furthermore, geological studies suggest the possible existence of substantial subsurface reserves of elemental sulfur on Mars [59]. Such a resource would significantly simplify the process bypassing the need for complex chemical synthesis, thus reinforcing sulfur's role as a viable

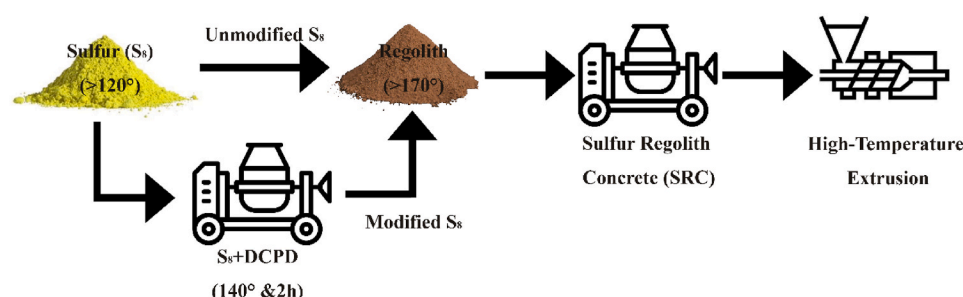


Fig. 5. Preparation, mixing, and extrusion of Marscrete. Adapted from: Giwa et al. [53].

binder for Martian construction. The production of sulfur-based mar-screte involves heating sulfur to 120 °C and mixing it with Martian regolith preheated to approximately 170 °C for 2 h. Dicyclopentadiene (DCPD) can also be added to the sulfur to enhance the mechanical and thermal performance of the resulting composite [60]. Due to the planet's low atmospheric pressure, entrapped air is negligible, eliminating the need for vibration during casting. However, at pressures below the triple point of sulfur (~0.014 atm), sulfur does not form a stable liquid and tends to sublime directly from solid to gas. Therefore, to achieve a workable molten state, the process requires either insulated or pres-surized robotic systems capable of maintaining conditions above the triple point during heating and mixing.

Regarding construction methods, structural components can be manufactured using high-temperature extrusion systems capable of printing or spraying the mixture, see Fig. 5. Since Martian environ-mental conditions cannot naturally reach the required temperatures, the process must rely on insulated and pressurized robotic systems equipped with electrical or chemical heating.

3.4. Layout design: architectures, functionalities, structural design

The work of Gernhardt et al. [61] provides the basis for design guidelines for deep space habitats, balancing functional requirements with human and psychological factors. Although the Deep Space Habitability Design Guidelines [61] were initially developed for deep-space crewed missions, they can also be extended to the structural and architectural design of future Martian habitats. Well-organized spaces are prioritized over larger volumes, even though increased vol-ume could facilitate functional separation and a perception of reduced confinement. Habitats should incorporate redundancy in functional areas to enhance operational efficiency, support fault management, and optimize time use. In larger volumes, zone separation can be achieved through bridges, with the primary goal of minimizing cross-contamination between dirty and clean areas. We highlight that floors, walls, and pallets should be engineered to support this separation and facilitate continuous cleaning. Smooth, easily sanitized surfaces are preferred, as well as proper placement of systems for waste collection and management in reduced gravity (WCS) and wet-waste management systems, which must ensure isolation, ventilation, and incineration to prevent odor propagation. The habitat should also implement clear coding schemes for the organization of mission-critical objects, enabling the crew to access them quickly and intuitively. Finally, observation windows providing views of the Martian landscape, rendezvous events, and robotic operations are essential for the psychological well-being of the crew.

Regarding the structural design, once the constraints in terms of functionalities, geometries and material properties are set, the design will cover aspects related to structural integrity under specific loading cases. Due to the peculiar nature of this design phase, we refer the reader to Section 4.5 for the proper discussion of the analyzed case-study.

4. Case study: ERETHEA Mars habitat

In the following, the design process to develop a new Mars habitat called *Erethea*, is detailed.

4.1. Mission design: selection of mission type and landing site

For the selection of the mission type, reference was made to the NASA report [30], which after evaluating the pros and cons of both, endorsed the recommendation to pursue long-stay missions. The only considerations slightly favoring a short-stay mission were its lower cost and shorter duration, which, however, entail significant operational limitations. The recent CHAPEA mission [62], involving a one-year surface stay and also promoted by NASA, further confirms the agen-cy's preference for long-stay missions. Accordingly, for the purposes of

Table 3

Summary of the engineering constraints related to the Martian surface and terrain for the Perseverance rover, as defined in the landing site selection of NASA's Mars Perseverance mission, adapted for human landing site selec-tion. Adapted from: Pajola, M. et al. [50].

Engineering Factors	Data
Latitude	30°S to 30°N
Altitude	<+0.5 km
Landing ellipse diameter	100 km
Slope at 2–10 km scale	< 5.71°
Slope at 50 m scale	<10°
Slope at 18 m scale	< 20°
Rock abundance	K<12%, locally K<20%
Thermal inertia	>100 Jm ⁻² s ^{-0.5} K ⁻¹
Albedo	< 0.25

this study, a long-stay mission is selected. The optimal launch window for a long-stay mission to Mars occurs approximately every 26 months.

Regarding the landing site, the proposed location for our case study is an ancient area of Mars, within Arabia Terra, centered at 6°10'N and 354°540'E, called Vernal Crater [50,63]. Indeed, based on the work of Pajola et al. [50], the proposed landing site demonstrates a high degree of compatibility with both engineering and scientific requirements, as mentioned in Section 3. Table 3 summarizes the main features of the Vernal. This site is of particular interest because according to the mea-surements from the Mars Odyssey neutron spectrometer, hydrated minerals and/or subsurface ice seem abundant. Indeed, a water equiv-alent hydrogen (WEH) value of ~13.5 ± 0.6 wt% was recorded at Vernal Crater [50], among the highest recorded on the Martian surface besides the poles. The presence of water could serve as both a vital resource and a hydraulic binder in the production of regolith-based construction materials. The site, located in Arabia Terra, also holds strong scientific relevance. The surrounding Regions of Interest (ROI) include volcanic deposits and stratified sediments (Fig. 6), likely linked to an ancient basin. Noteworthy, the crater's morphology, marked in the northwest by extensive erosion that produces relatively gentle slopes, facilitates mobility from the landing point both outward toward volcanic deposits and inward toward layered sediments. This configuration makes the area especially suitable for future exploration missions, offering opera-tional safety alongside access to sites of significant geological and astrobiological interest.

4.2. Loads and actions for structural design and verification

The following loads and actions were considered: asteroid impacts; GCRs and SPEs radiations; seismic loads; and dead loads.

The first action worth being analyzed is the asteroid impacts case study. Fig. 7 shows an impact map obtained from the events catalogued by Daubar et al. [39] on the overall Martian surface: during the 13-years observation period the selected landing site [50] did not experience closely spaced or particularly intense impacts. This fact however does not assure that fresh impact events may have occurred before the 12.925 years analyzed timeframe by Daubar et al. [39], as shown in Pajola et al. [50], or may occur in the future.

The occurrence of impact events is modeled as a homogeneous Poisson process, consistently with the common assumption adopted in reliability analyses of meteoroid loading on extraterrestrial structures [64]. This modeling approach is appropriate for rare, spatially and temporally independent events occurring with an approximately con-stant mean rate over the observation period, conditions that reasonably characterize impact phenomena at the scale considered in this study. To compute the probability of an impact we consider an area of 100 m² where the habitat is located, centered on the landing site. The average impact density per unit area within the observation period can be calculated (Eq. (2)) as:

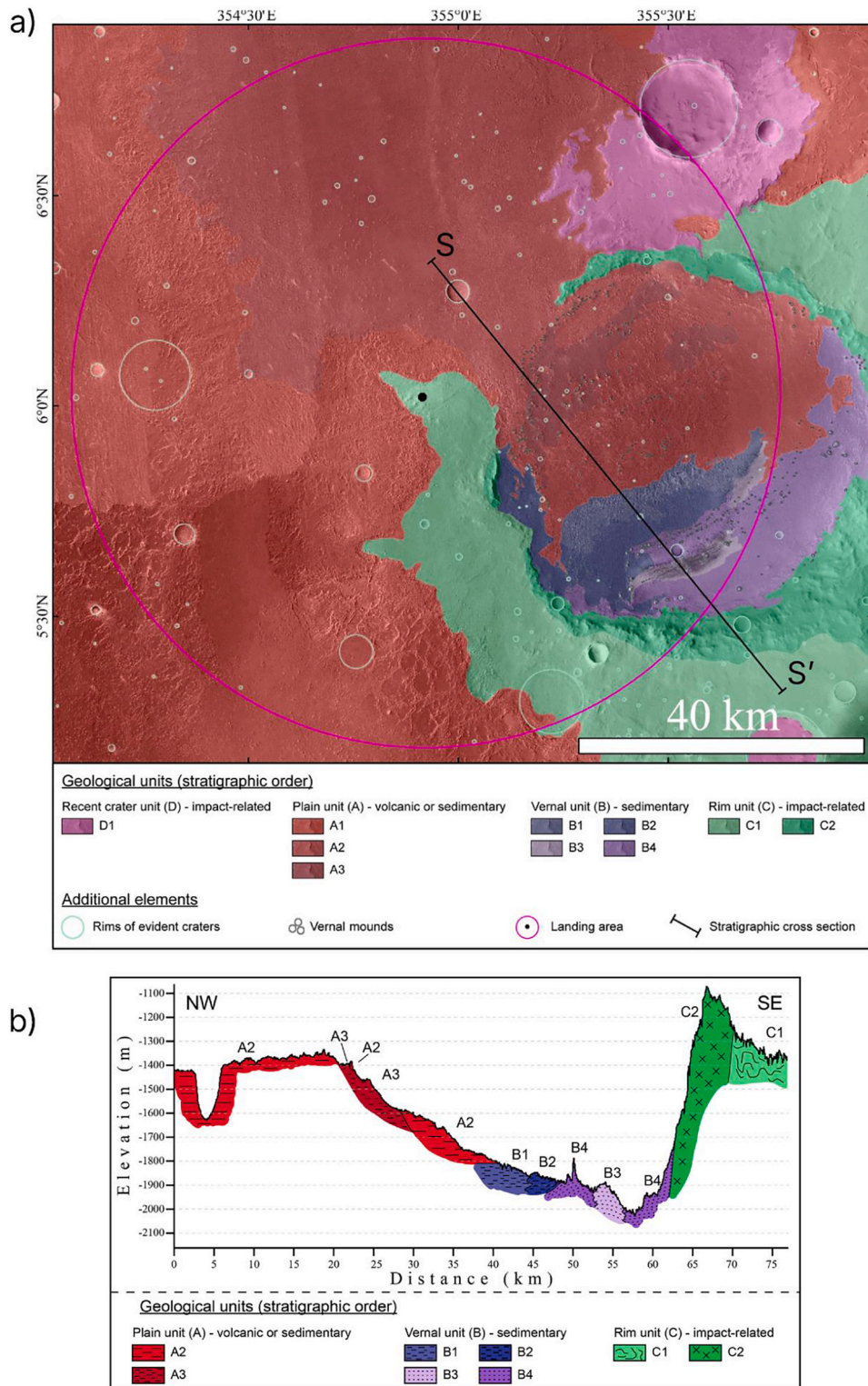


Fig. 6. a) The geological map shows the proposed landing site and its immediate surroundings, with each unit described in detail in Pajola et al. [50]. The center of the landing circle is marked by a black dot, and the black line indicates the location of the stratigraphic cross-section (ss) b). Adapted from: Pajola et al. [50].

$$\lambda = \frac{\text{number of events}}{\text{observation period} \cdot \text{surface area}} \quad (2)$$

where the number of events is 1203, the observation period is 12.925 years [40], and the total surface area of Mars is $1.44 \times 10^8 \text{ km}^2$. Assuming that impact events can be modeled as a homogeneous Poisson process, the number of events $N(A, t)$ occurring over an area A during a

time interval t follows a Poisson distribution with mean $\mu = \lambda At$, where μ represents the expected number of impact events, and λ is the mean impact rate per unit area and per unit time. Considering a reference area of 100 km^2 for a macro-hazard assessment around the landing site, the probability of observing at least one impact event within one year is approximately 0.01%. If we consider as duration a theoretical service life of 50 years, according to Refs. [64,65], and the real footprint of the

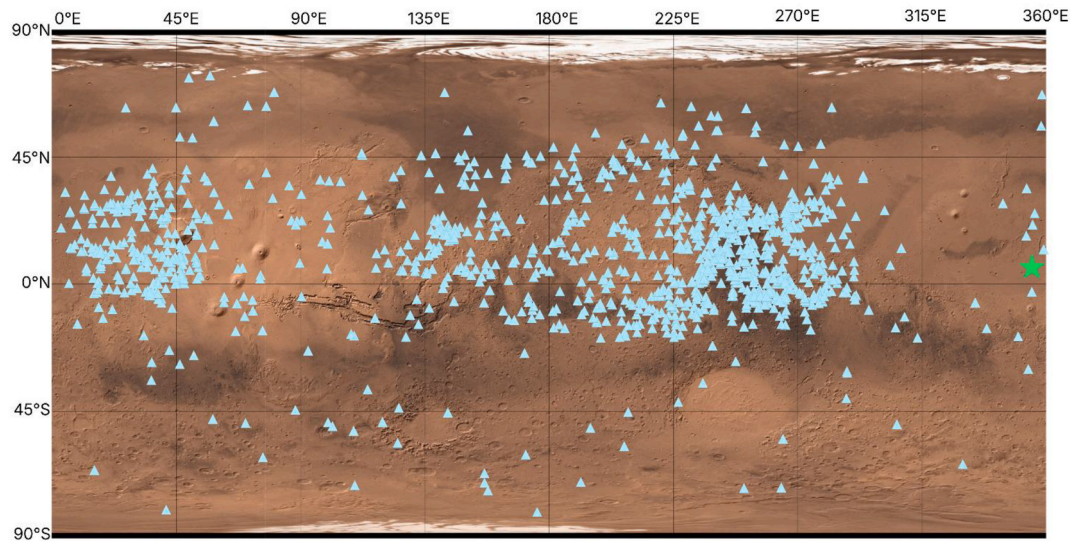


Fig. 7. Impact map of Mars. Blue triangles indicate impacts, while the star represents the designated landing site on Vernal crater. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

construction habitat itself, e.g. 100 m^2 , the probability of having an impact at the structure is $6.4 \cdot 10^{-12}$. Accordingly, we assume at this stage of the work that a major impact event (i.e., an event creating a crater such as those classified by Daubar et al. [39]) is not a design-driving action. Any small-scale impacts (i.e., creating craters with few centimeters in diameter) will be addressed primarily through the general robustness of the structure itself.

Regarding GCRs and SPEs radiations, the design should take into account their presence in order to guarantee sufficient sheltering to the astronauts. In practice, the habitat should be designed with a surrounding material that attenuates the effect of the radiation below a certain limit. This limit is properly defined in NASA STD 3001 [66], which specifies that the cumulative effective radiation dose for an individual crew member over their entire career must not exceed 600 mSv, regardless of the age and sex. Effective exposure calculations must account for the time spent inside the habitat, the shielding efficiency, and the frequency of extravehicular activities. During solar minimum conditions, SPE radiations decreases significantly, while GCRs increase; however, SPEs typically last only a few hours or, at most, days [40]. During these events, crew members should shelter in areas of the habitat

or vehicle with enhanced shielding. It is further assumed that crew members spend two-thirds of their time in a lightly shielded area and one-third of the time sleeping in a heavily shielded area [30]. With these precautions, the total dose to the most radiation-sensitive organs, the hematopoietic system, is estimated to be between 395 and 450 mSv. To account for radiation in the design, it is thus necessary to evaluate a minimum thickness of the sheltering layer present in the habitat. Conservatively, considering a habitat made with Martian regolith only, the minimum thickness for adequate shielding, can be calculated through the Bethe formula (Eq. (3)) to calculate radiation attenuation effects [67–73]:

$$\frac{dE}{dx} = \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \frac{4\pi z^2 N_0 Z \rho}{m_e c^2 \beta^2 A} \left(\ln \left(\frac{2m_e c^2 \beta^2}{I_{av}} \right) - \ln(1 - \beta^2) - \beta^2 \right) \quad (3)$$

where dE/dx is the amount of energy transmitted through the material as a function of its thickness, Z , A , and ρ denote, respectively, the atomic number, atomic weight, and density of the stopping material. ϵ_0 is the permittivity of free space, and I_{av} is the mean excitation energy of the stopping material, expressed in electronvolts (eV). N_0 is Avogadro's

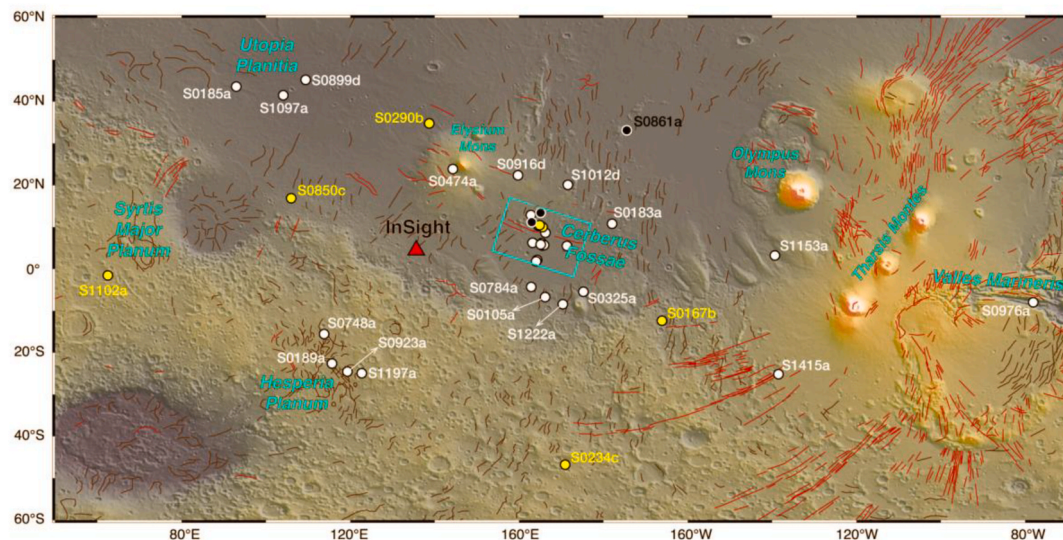


Fig. 8. Spatial distribution of low-frequency marsquakes. Adapted from: Wang et al. [44].

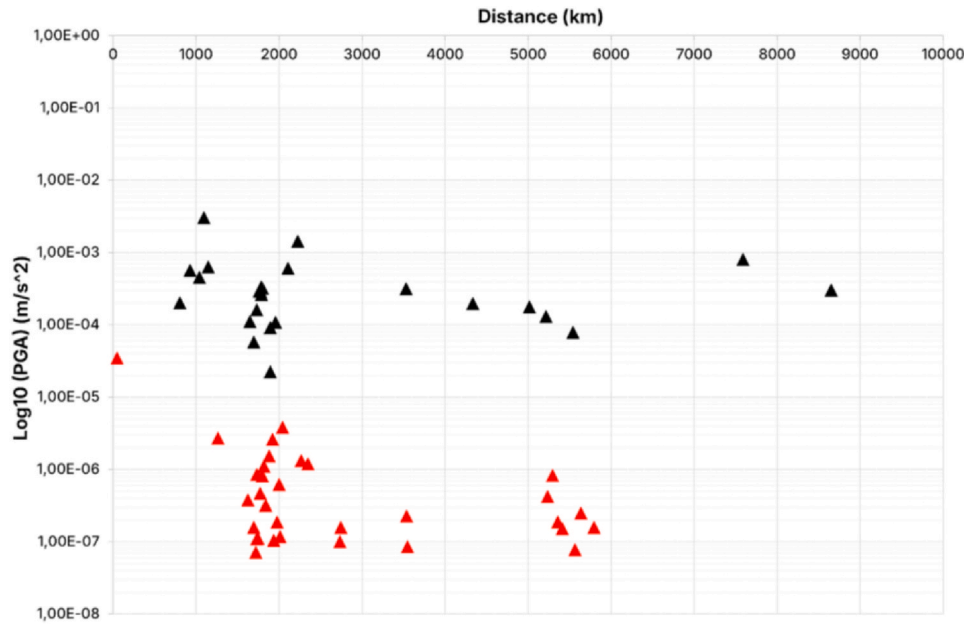


Fig. 9. Initial 58 analyzed events: black denotes events recorded at 100 Hz, and red denotes those recorded at 20 Hz (later removed in the final dataset). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Table 4
Subset of 21 marsquakes for the calibration of the attenuation law.

Event	Magnitude (M_w)	Distance (km)	PGA (m/s^2)
S1222a	4.6	2223.7	1.43E-03
S0976b	4.1	928.5	5.65E-04
S1094 b M	4.0	3530.6	3.15E-04
S1157a	3.8	2105.4	6.04E-04
S1133c	3.8	1786.0	2.61E-04
S0421b	3.3	5541.4	7.81E-05
S0809a	3.3	1762.4	2.92E-04
S0820a	3.3	1786.0	3.33E-04
S1337a	3.3	1147.3	6.31E-04
S1415a	3.3	5216.1	1.30E-04
S1102a	3.2	4335.0	1.97E-04
S0794a	3.1	1040.9	4.52E-04
S0864a	3.1	1697.3	5.71E-05
S0918a	3.1	1650.0	1.10E-04
S0923a	3.0	1957.5	1.07E-04
S1039b	3.0	1797.8	3.22E-04
S1153a	3.0	5015.1	1.76E-04
S1223a	2.9	1094.1	3.03E-03
S0916d	2.9	1732.8	1.60E-04
S0802a	2.9	1774.2	2.89E-04
S0861a	2.9	1892.5	9.10E-05

constant; k_0 is the Coulomb constant; m_e is the electron rest mass; c is the speed of light in vacuum; z is the atomic number of the incident particle; and e is the magnitude of the electron charge. The parameter β is defined as the ratio of the particle's velocity to the speed of light, where the particle velocity is computed as a function of its initial (relativistic) kinetic energy T , this function of $m_{p/\alpha}$ (the radiation particle's rest mass), γ is the Lorentz factor, and c [67].

Another important action has to be considered for the habitat design, and this is the seismic action. Despite the landing site being located far from the Cerberus Fossae area affected by the recurrence of seismic events as shown in Fig. 8, we have decided to conservatively consider the presence of this action. To do so, we have reviewed the available data collected from the InSight catalog [43,74,75], and formed a subset of relevant seismic data. Then, on this subset an attenuation law of relatively similar events (i.e., within a fixed range of magnitude) has been calibrated. This law correlates an intensity measure of the quake

Table 5
Terrestrial ground motion attenuation laws and goodness-of-fit indicators.

Empirical Law	R^2	Adj R^2	RSME
Musson et al. (1994) [80]	0.385	0.317	0.801
Radu et al. (1994) [81], Lungu et al. (1995a) [82] & Lungu et al. (1996) [83]	0.378	0.268	0.829
Molas & Yamazaki (1995) [84]	0.406	0.258	0.362
Singh et al. (1996) [85]	0.372	0.339	0.342
Baag et al. (1998) [86]	0.378	0.309	0.350
Reyes (1998) [87]	0.461	0.326	0.795
Chapman (1999) [88]	0.461	0.326	0.345
Wang et al. (1999) [89]	0.406	0.302	0.351
Xiang & Gao (1994) [90]	0.303	0.226	0.058
Theodulidis & Papazachos (1992) [91]	0.378	0.309	0.806
Garcia-Fernández & Canas (1991) [92] & Garcia-Fernandez & Canas (1995) [93]	0.363	0.292	0.354
Sen (1990) [94]	0.376	0.307	0.806
Fukushima et al. (1988) [95] & Fukushima & Tanaka (1990) [96]	0.385	0.277	0.358
Singh et al. (1987) [97]	0.378	0.309	0.350
Krinitzsky et al. (1987) [98] & Krinitzsky et al. (1988) [99]	0.385	0.317	0.347

with an epicentral distance [76], and its use will allow a designer to consider e.g. a certain PGA consistently with a hypothetical distance from the event origin. First of all, consistent with the analyzed dataset [74,75], a hypocentral depth of 50 km is assumed. Other studies support similar values: Drilleau et al. [77], analyzing 20 significant seismic events, found an average depth of approximately 40 km, in agreement with Brinkman et al. [78] and Durán et al. [79]. Shallower depths (0–10 km) are considered unlikely, as high-frequency fundamental-mode surface waves from deeper events would fall below the instrumental noise (<0.1 Hz) [77]. Then, an initial subset of 58 events were selected from the available catalogues [43,74,75], ranging from 2.7 to 4.6 M_w , see Fig. 9. Analyzed data include velocity and acceleration components (U , V , W), back-azimuth, epicentral distance, and data quality. Velocity derivatives with respect to time were used to determine PGA values, scaled and calibrated according to the factors provided in the dataset. Events were also categorized based on the sampling interval ($\Delta t = 0.01$ or 0.05 s). To prevent any loss in terms of data regarding

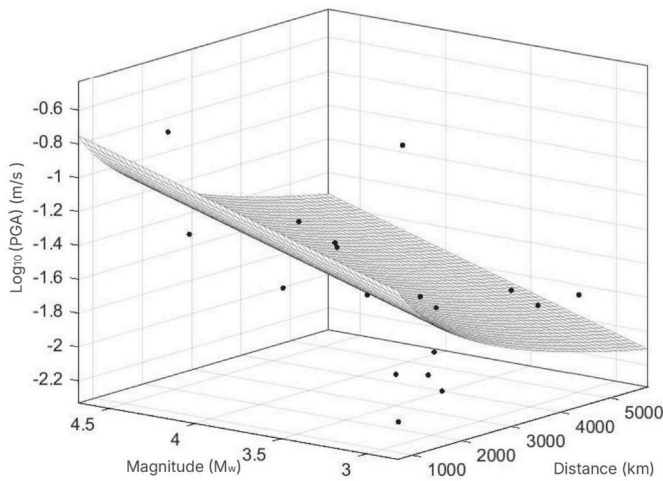


Fig. 10. Attenuation law by Singh et al. (1996) [97] and its application to the marsquake dataset.

acceleration peaks due to excessively large time intervals, only events recorded at 100 Hz were selected. Thus, from this initial subset, only 21 events met the established criteria and are reported in Table 4, which have magnitudes ranging from 2.9 to 4.6. Hence, it is now possible to define a mathematical relationship between *PGA*, magnitude, and epicentral distance for the Martian context, which is known as ground motion attenuation law.

Initially, 15 terrestrial ground motion attenuation laws were applied to the selected dataset, which are listed in Table 5 together with goodness-of-fit indicators. This selection was carried out considering formulations that include only parameters such as magnitude, *PGA* and distance, excluding factors related to soil characterization or shear wave velocity, which are unavailable for Martian data. Further, they should not depend on Earth's gravitational acceleration, which is not representative for the Martian context. Results indicate relatively good coefficients of determination, ranging between 0.3 and 0.6, for all the models. Considering the very limited dimension of the dataset and data scarcity, results are considered very promising. However, not all the relationships correctly represent the expected trend: for instance, the formulations by Reyes [87], Chapman [88], and Xiang & Gao [90] exhibit increasing slope with decreasing magnitude and increasing epicentral distance. Hence, the most suitable attenuation law for the marsquakes dataset is found to be Singh et al. [97] relation, whose validation on the marsquake data is shown in Fig. 10. Its formulation is represented in Eq. (4), where *PGA* is expressed in cm/s^2 ; *M* is the moment magnitude of the event; *D* is the epicentral distance expressed in kilometers, and the calibration coefficients are $a = 0.1987$; $b = 0.3100$; $c = -0.8448$.

$$\log_{10}(PGA) = a + bM + c\log_{10}(D) \quad (4)$$

Finally, temperature variations were found to be non-negligible, thus it is expected that they will induce relevant effects on the habitat design. Considering the thermal load and its interaction with other structural loads is essential for realistically assessing the overall response under extreme environmental conditions. Moreover, evaluating thermal action not only allows for the interpretation of stresses and deformations but also enables an assessment of the material's potential insulation properties and the degree to which the internal temperature depends on external conditions. To evaluate thermal effects, time-dependent temperatures reported in Table 2 were applied on the external surface of the structure, while maintaining a constant inner temperature, creating a through-thickness thermal gradient. Regardless of the structure geometry, in theory, this gradient compresses the upper fibers and stretches the lower ones, generating concentrated nodal moments within the

Table 6

Summary of environmental effects and their influence on habitat design.

Environmental Effect	Consideration in Design	Reasoning
Asteroid Impacts	Not design-driving	Probability of significant impact over 50 years is extremely low ($p \approx 6.4 \cdot 10^{-12}$ for habitat footprint), negligible dynamic loading, and structural robustness suffices for minor micrometeoroid events
Wind	Not design-driving	Martian atmospheric density ($\sim 1.63\%$ of Earth's atmospheric density) produces negligible aerodynamic forces, insufficient to induce structural stresses or deformations
Gravity Loads	Design-driving	Constant vertical load governs structural sizing, foundation design, and material strength requirements
Seismic Activity	Design-driving	Marsquake-induced accelerations and <i>PGA</i> are used to evaluate dynamic response, ensure stability, and define peak load combinations
Temperature Variations	Design-driving	Thermal gradients induce expansion/contraction, bending moments, and internal stresses; critical for material selection and structural detailing
Radiation (GCRs, SPEs)	Design-driving	Shielding must reduce cumulative dose below NASA STD 3001 limits [66]; material thickness and composition directly affect crew safety and habitat layout

structure.

Table 6 summarizes the environmental effects considered in this study and their influence on the habitat design, distinguishing between factors that are design-driving and those that can be conservatively disregarded.

4.3. Sulfurcrete as a marscrete candidate

The selected material is a sulfurcrete, whose realization, casting and application potential has been already investigated in Section 3. The study by Wan et al. [52] was selected as a reference for identifying key properties of sulfur marscrete relevant for the structural design of the habitat. In their work, molten sulfur was mixed with the JSC Mars-1A Martian regolith simulant at sulfur contents ranging from 35% to 60% by weight, blended above 120°C , and cast into molds before cooling. Two aggregate sizes were tested (≤ 1 mm and up to 5 mm), with smaller aggregates providing more homogeneous samples and fewer voids. Mechanical characterization included unconfined compression, three-point bending (notched and un-notched), and indirect tensile (Brazilian) tests performed after 24 h. The optimal sulfur content was identified as 50% ($\pm 2.5\%$), at which cubic specimens reached compressive strengths of 48 MPa, increasing to 58–63 MPa after annealing for the 1 mm aggregate mixtures. Stress-strain curves for mixtures with 40–60% sulfur confirmed this optimum. Three-point bending tests yielded nominal flexural strengths of 1.65 MPa, increasing to 2.3 MPa after annealing, with un-notched specimens showing an average Modulus of Rupture of 7.24 MPa. Indirect tensile tests on annealed cubic samples with sulfur contents of 47.5%, 50%, and 52.5% produced splitting tensile strengths of $3.6 \text{ MPa} \pm 30\%$, $3.9 \text{ MPa} \pm 28\%$, and $2.72 \text{ MPa} \pm 26\%$, respectively, again confirming the 50% mixture as optimal. Numerical simulations using a Lattice Discrete Particle Model (LDPM), in the study of Wan et al. [52], supported these results and estimated a fracture energy of 67 J/m^2 , with predicted compressive strengths around 54 MPa. Overall, these findings provide the essential material parameters required for structural design considerations in the present work. Furthermore, regarding the radiation shielding properties of this material, the combination of sulfur and Martian regolith provides modest additional protection compared to

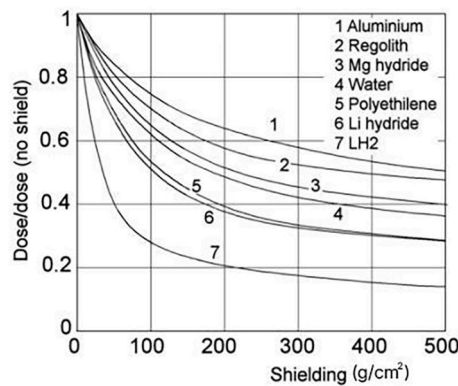


Fig. 11. Absorbed dose as a function of shield thickness for different shielding materials. Adapted from: Genta [68].

regolith alone. While sulfur has a relatively high material density, it does not significantly increase the nucleus packing density, which is the primary factor determining resistance to Galactic Cosmic Rays (GCR). As shown in Fig. 11, unprocessed regolith alone can attenuate nearly 50% of the radiation at an areal density of 300 g/cm² [68]. The incorporation of sulfur is primarily dictated by structural considerations: pure regolith lacks cohesion and cannot form rigid, load-bearing elements, whereas sulfur acts as a binder, producing a solid, self-supporting material suitable for habitat construction. Therefore, any marginal improvement in radiation shielding is a secondary effect, and the overall shielding efficiency remains fundamentally limited by the low hydrogen content of the mixture.

A minimum thickness of approximately 1.5 m, so, is sufficient to reduce exposure to GCRs and SPEs below 50 mSv per year, limiting total exposure during a 545-day mission to roughly 75 mSv, which is not harmful to crew health [30]. Hydrogen-based materials, such as hydrogenated polyethylene, water, or metallic hydrides, remain valuable choices for further shielding layers, but are not suitable for constructing the structural elements. Thus, radiation protection can be further improved by adding inflatable structures (Fig. 12 [100]) inside the habitat, which act as an extra protective layer, reducing the crew's exposure to radiation. Additionally, the proposed minimum thickness of 1.5 m represents only the exposed portion of the structure, as some

sections will be partially buried. In deeper areas, shielding may be augmented by layers of regolith ranging from a few centimeters to several meters. The study conservatively considers only the marscrete thickness, thus underestimating the overall shielding potential of partially buried sections.

The structural model is therefore developed assuming the thickness of 1.5 m as a fixed geometric parameter, while the structural analyses aim to verify the adequacy of the design derived from the minimum thickness required for radiation shielding.

Summarizing, the main material properties of the sulfurcrete are listed in Table 7.

4.4. Spaces, volumes and architecture

The main design concept on which this habitat relies is the coupling of an external structure, which has the main role to be self-standing and protect the crew from radiations and external actions, and an internal structure, that allows for spaces and volumes definition. The external structure made with the sulfurcrete is the object of the specific structural design, while the internal structure is made of inflatable modules which define the final geometry of the habitat.

Starting from the internal modules, deployable structures for Martian habitats represent the most logistically efficient solution. The primary advantage of this approach is the maximization of internal volume once deployed, compared to the compact, stowed configuration required during transport aboard cargo spacecraft. This modularity also offers weight benefits, as modules can be autonomously anchored to the ground and subsequently covered with regolith-based concrete shielding, thereby avoiding additional launch loads. Key advantages of these structures include:

- additional radiation protection provided by polymeric materials;
- modularity allowing multiple habitat configurations and adaptability during the mission;
- relatively large internal volumes, optimizing habitability;
- integration of systems for lighting, pressurization, and waste management within the wall thickness;
- elevation of the module above the ground, mitigating thermal effects from the soil;
- rapid and simple connection between modules;

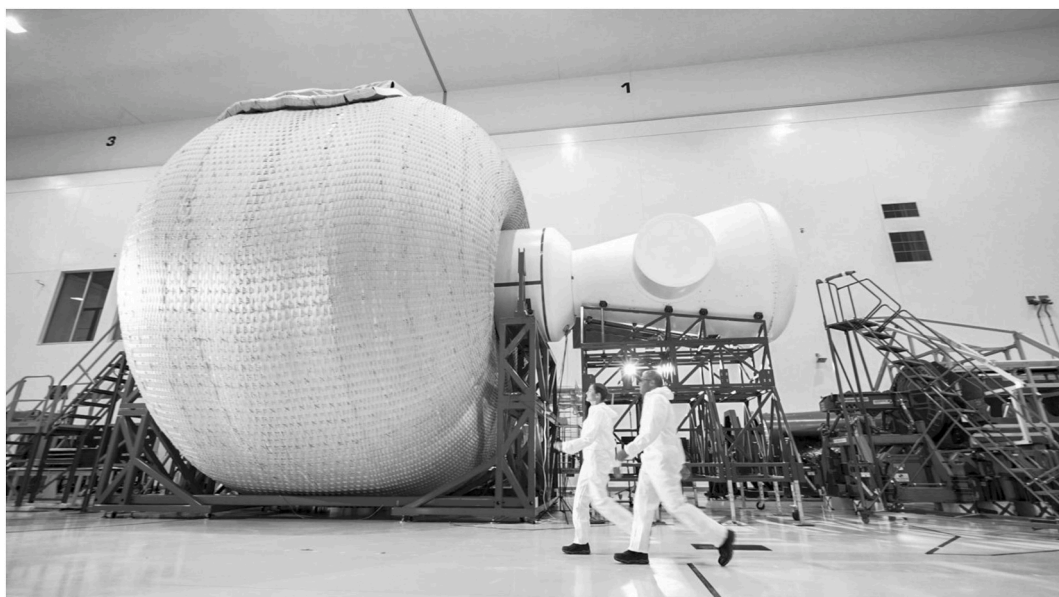


Fig. 12. Inflatable habitat models proposed for interplanetary missions. Adapted from ATA (2023) [100].

Table 7
Summary of the material properties of the sulfurcrete.

Mix Design (weight %) [52]	Sulfur-based binder with a density of 1.7 g/cm ³ Martian regolith-based aggregate with a density of 2.07 g/cm ³	Sulfur (S8)	50.0
		Silicon dioxide (SiO ₂)	34.5–44
		Titanium dioxide(TiO ₂)	3–4
		Aluminum oxide (Al ₂ O ₃)	18.5–23.5
		Ferric oxide (Fe ₂ O ₃)	9–12
		Ferrous oxide (FeO)	2.5–3.5
		Magnesium oxide (MgO)	2.5–3.5
		Calcium oxide (CaO)	5–6
		Sodium oxide (Na ₂ O)	2–2.5
		Potassium oxide (K ₂ O)	0.5–0.6
		Manganese oxide (MnO)	0.2–0.3
		Diphosphorus pentoxide (P ₂ O ₅)	0.7–0.9
	Rock aggregate with a maximum size of 1 mm.		
Strength (MPa)	Compression	48	[52]
	Bending	7.24	[52]
	Splitting	3.9	[52]
Mixing and setting time	Mixing temperature	145 °C	[53]
	Post-extrusion/spraying setting time	24 h at room temperature, significantly faster at low temperatures like those on Mars.	[52]
Rheological properties	Young Module (GPa)	10	[52]
	Poisson's ratio	0.18	[52]
	Fracture Energy (J/m ²)	67	[52]
	Dynamic yield stress (Pa a 145 °C)	400	[53]
	Static yield stress (Pa a 145 °C)	530	[53]
	Plastic viscosity (Pa.s a 145 °C)	10.5	[53]
	Fresh density (g/cm ³)	2.310	[53]
	Hardened density (g/cm ³)	≈2.1–2.3	[53]
	Interlayer adhesion	Strength is high at 145 °C and optimal at 150 °C. Preheating the substrate to 85 °C increases strength by 113% compared to 55 °C.	[53]

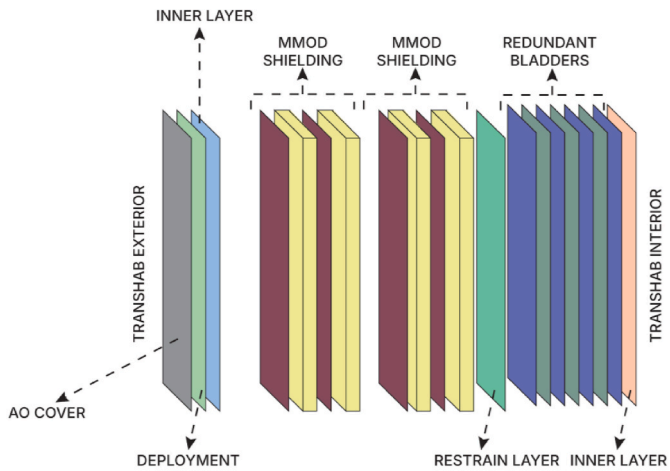


Fig. 13. The stratigraphy of the architectural assembly of the inflatable structure. Adapted from Valle et al. [101].

- capability to safely withstand the operational internal–external pressure differential (≈1 bar) through multilayer softgoods architectures, including structural restraint layers and redundant bladder

systems, enabling reliable pressurized operation independent of the external protective shell [101].

The stratigraphy [101] shown in Fig. 13 of these inflatable modules consists of a total thickness of 40 cm, divided into five main layers:

1. AO Cover (Atomic Oxygen Cover): external protection against highly erosive atomic oxygen, typically made of durable materials such as Kapton or coated Teflon;
2. MLI Layer (Multi-Layer Insulation): multilayer thermal insulation with reflective sheets, Mylar, and Dacron interleaved with spacers, protecting against extreme temperatures in space;
3. MMOD Shielding (Micro-Meteoroid and Orbital Debris Shielding): protective layers originally designed to withstand high-velocity micrometeoroid and orbital debris impacts; in the context of the proposed Martian habitat, this shielding is effective against potential secondary fragments, such as chips or debris from the sulfurcrete structure itself;
4. Redundant Bladders: multiple sealed membranes maintaining pressurization even if one layer is compromised;
5. Restraint Layer and Inner Liner: textile containment structure and internal lining, safe, fire-resistant, and compatible with human occupancy.

Additionally, to address acoustic and psychological factors, a variable thickness (1–3 mm) acoustic polymer film [102] is planned to be integrated within the stratigraphy. This layer is intended to reduce acoustic focusing and echoes generated by the hemispherical dome geometry, thereby improving acoustic hygiene and mitigating infrasound, supporting crew comfort and well-being in the confined habitat environment.

Once the stratigraphy of the inner structure is defined, the spatial organization of the project should be designed, based on a clear functional articulation of environments, aimed at ensuring operational efficiency, sanitary safety, and users' well-being. For such scope, we select three main modules, here called *Erethea Module* (EM), anchored to the ground, designed to both separate and logically connect different activities: EM1: 3 × 3 × 3 m; EM2: 5 × 5 × 5 m; EM3: 7 × 7 × 7 m.

Although the layout is to be understood as notional and representative rather than fully-designed, the dimensions of the Erethea Modules (EM1–EM2–EM3) were determined based on functional, ergonomic, and structural considerations, with EM1 serving primarily as a transition and sanitation zone, where compact dimensions are sufficient to ensure controlled circulation and contamination mitigation between compartments while allowing seamless integration with the other modules and effective coverage by the external sulfurcrete shell. On the Martian surface, in fact, contamination risks arise from multiple sources, including the intrusion of fine dust containing chemically reactive compounds (e.g., perchlorates), particulate transfer from EVA and airlock operations [103], microbial transport between hygiene, medical, and living areas, and localized chemical or particulate releases that may affect ECLSS performance [30]. In closed-loop environments, contaminant accumulation can impact crew health, air quality, and system reliability; therefore, functional zoning and controlled circulation are adopted to limit contaminant propagation and maintain long-term environmental stability. EM2 accommodates daily activities, service areas, and laboratories, requiring larger volumes to allow comfortable occupancy, equipment placement, and efficient operational workflows, providing adequate ergonomic space, ventilation, and user safety. EM3 is reserved for specialized or private functions, such as medical support, conference rooms, and private accommodations, where high spatial quality and flexibility are required, enabling flexible internal configurations and optimal comfort. These dimensions follow a progressive modular logic, where each module increases according to the complexity and number of activities it hosts, ensuring a balance between functionality, comfort, ergonomics, modularity, and construction

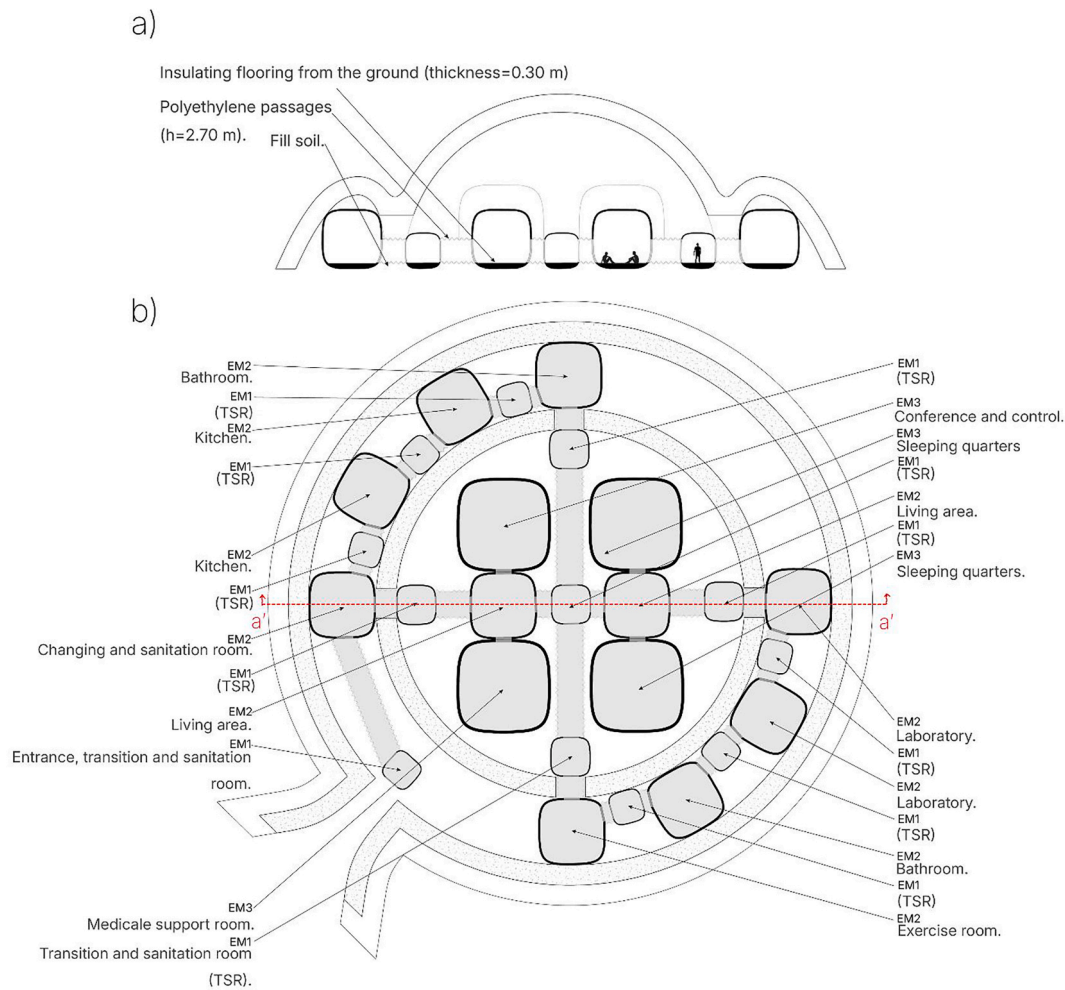


Fig. 14. a) Architectural plan Section (a'-a') of the habitat and b) Architectural plan.

feasibility in a Martian habitat context while maintaining coherence with the external shell and connecting corridors.

In parallel with the spatial layout, lighting design is critical to support crew well-being during long-term habitation. The internal surfaces of the inflatable modules should be complemented by a lighting system that ensures uniform illumination, minimizes shadows and glare, and integrates biofotonically-tuned spectral cycles. Such an approach can reinforce circadian entrainment, optimize visual comfort, and reduce eye strain, while supporting operational tasks across different functional areas. Techniques may include indirect or diffused lighting, spectral modulation throughout the day, and localized task lighting within the modules. Research has shown that short-wavelength light strongly influences human melatonin levels, alertness, and thermoregulation [104], and that blue-enriched white light can improve self-reported alertness, performance, and sleep quality in work environments [105], underlining the importance of spectral tuning in long-duration, windowless habitats. Regarding the external structure, architecturally, the layout is organized around a central dome and a surrounding toroidal ring. The overall form, characterized by curved and symmetrical volumes, conveys efficiency, adaptability, and protection, referencing archetypal models such as geodesic domes or orbiting space stations.

Globally, each environment fits into a coherent network where form, function, and structure reinforce one another. Modular units, autonomous yet interconnected, standardizable, and prefabricable, make the structure suitable for remote or temporary contexts, optimizing construction time, logistics, management, and maintenance. The project,

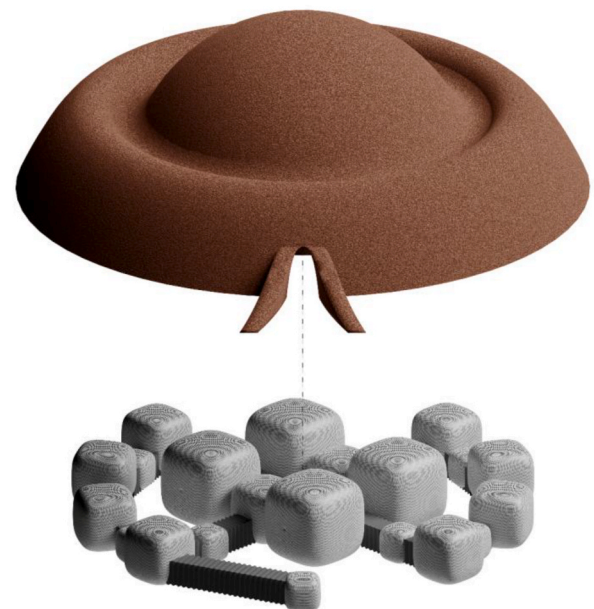


Fig. 15. Axonometric exploded view of the project.

therefore, constitutes a replicable and connectable platform capable of evolving over time with the users' needs. Fig. 14 shows a final view of

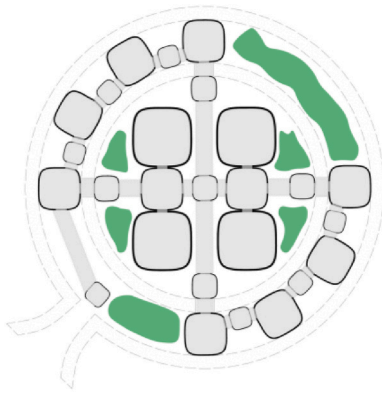


Fig. 16. Areas in green designated for future cultivation or specific zones for waste disposal. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

the proposed project, with a) the architectural plan, and b) the section, while Fig. 15 shows an axonometric exploded view of the habitat. From a dimensional perspective, the structure features a compact yet spacious configuration, with an overall external diameter of approximately 25.8 m. Within this circumference, spaces are organized into radially distributed modules, with corridors ranging from 1.5 to 2.2 m in width, sufficient to ensure both safe personnel passage and integration of building systems. The largest functional units, such as lounges and conference rooms, reach internal widths of approximately 7 m, allowing flexible use of space without compromising compartmentalization. These proportions are carefully calibrated to meet functional, ergonomic, and structural requirements, ensuring a balance between internal spaciousness and external compactness - essential in isolated environments requiring autonomy and energy efficiency.

A key feature of the project is its high degree of modularity and adaptability. The combination of a regular geometric form (central dome and peripheral torus) and modular functional distribution allows for rapid reconfiguration of space usage and functional allocation in response to technological upgrades, functional changes, or structural expansions. Special attention has been given to residual spaces within the distribution mesh, particularly at intersections between main modules and between the dome and torus. Although these areas are not initially assigned specific functions, they are designed with flexible use in mind and can be converted into experimental or productive zones (Fig. 16), such as protected greenhouses for plant cultivation, supporting food self-sufficiency and the recycling of air and humidity.

4.4.1. Habitat environmental and safety systems

The habitat is designed as a closed-loop, controlled environment where air quality, particulate containment, and crew safety are primary design priorities, particularly to protect against the fine, abrasive, and potentially toxic Martian dust. [106]. The internal environment follows contamination-controlled criteria similar to cleanroom standards [107] [108], with an HVAC system featuring multi-stage HEPA/ULPA filtration, positive pressurization relative to the external environment, controlled air distribution and recirculation, high air exchange rates, and functional compartmentalization via pressure gradients to reduce cross-contamination risks [30]. Crew ingress and egress are managed through a multi-stage airlock system, for whose detailed design we refer to Vrankar et al. [109], including an external interface chamber for initial depressurization/pressurization, an intermediate decontamination chamber equipped with active dust mitigation systems (air showers, mechanical brushing, or electrostatic/vibration-based removal), and an internal buffer zone at controlled pressure, ensuring unidirectional, filtered airflow that progressively removes contaminants. The interior is organized into functional zones with increasing levels of environmental control, including Safe Havens (identified in EM1s) -compartments with

independent structural and pressure boundaries, autonomous life-support (air supply, CO₂ removal, thermal control), redundant sealing systems, and strategic placement to maximize survival in emergency scenarios [110].

4.5. Erethea structures

4.5.1. Design and verification under static and seismic actions

The geometry of the structure was modeled for subsequent import into Strand7 [111], a structural analysis software. The shape follows a catenary curve, selected for its structural and formal properties, distributed along two concentric ring profiles. The dome is a symmetric surface of revolution, rotated around a vertical axis. The entire process followed a parametric logic, allowing smooth modifications of the form based on the base curves. Overall, the modeled structure closely replicates the previously presented plans and sections, with one necessary adjustment: the intersection between the semi-torus (derived from the catenary) and the dome initially resulted in overlapping elements, which would have caused the structural analysis software to incorrectly account for double thickness at that location. To address this issue, a single element was introduced to connect both structures to the ground, ensuring correct interpretation in the structural analysis software. The geometry was then discretized in finite element software into 2181 quadrilateral plate elements for the analysis. Each element represents approximately 0.3% of the total area, as calculated by the software. Boundary conditions at the base in the FE model were applied as fixed supports.

The material assigned to the structure reflects the properties previously described in Table 7.

Regarding actions, two main types of loads were considered: self-weight and seismic action. The self-weight was defined using Martian gravity, 3.73 m/s², applied vertically downward. The seismic action was determined using the derived attenuation law (Eq. (4)) to find equivalent horizontal forces in the two directions, by setting the magnitude and distance parameters, which were fixed arbitrarily considering the most severe event ever registered on Mars as a lower boundary occurring close to the landing site. Thus, the magnitude was fixed at 6.0 M_w and the distance at 1 km, resulting in a PGA of 1.14 m/s² (0.31 Martian g); it was, then, modeled based on the accelerations determined in Ref. [112], decomposing into two orthogonal horizontal directions and weighted by a coefficient ψ of 1 along the primary direction (Eq. (5)) and 0.3 along the orthogonal direction (Eq. (6)), according to usual seismic design prescriptions present in many building codes [113,114]:

$$\text{Primary acceleration } (a_p) = 0.31 \cdot g \left(= 3.73 \frac{m}{s^2} \right) \cdot \psi \left(= 1 \right) = 1.156 \frac{m}{s^2} \quad (5)$$

$$\text{Secondary acceleration } (a_s) = 0.31 \cdot g \left(= 3.73 \frac{m}{s^2} \right) \cdot \psi \left(= 0.3 \right) = 0.347 \frac{m}{s^2} \quad (6)$$

In this study, the main direction is considered along the x-axis, with y-treated as the secondary direction. Due to the symmetry of the structure, it is not necessary to analyze other directions. Load combinations were defined according to common prescriptions present in many building codes, e.g. Refs. [112–114].

Specifically, regarding the seismic combination for ultimate limit states, it reads (Eq. (7)):

$$E + G_{k1} + G_{k2} + \psi_{2i} \cdot Q_{ki} + \psi_{2(i+1)} \cdot Q_{k(i+1)} \dots \quad (7)$$

where G_{k1} and G_{k2} represent the permanent structural loads; Q_{ki} denotes the characteristic variable actions; ψ_{2i} are combination coefficients, depending on the nature of the variable load; E represents the seismic action, i.e., the effect of the forces generated by the quake, calculated as the vector product of the two previously defined accelerations (see Eqs.

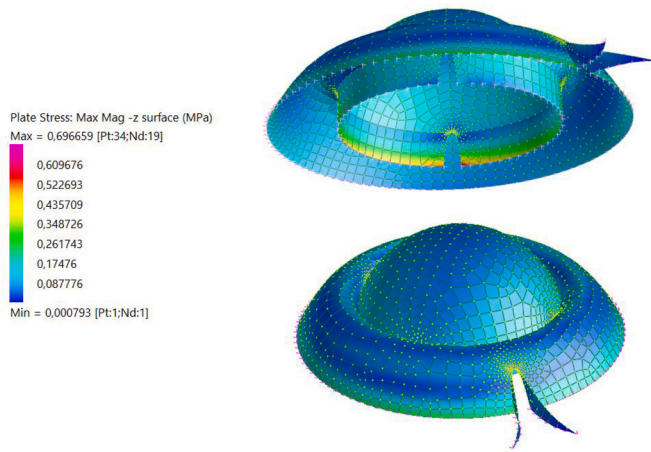


Fig. 17. Representation of the maximum principal stresses between directions 11 (maximum tension) and 22 (maximum compression).

(6) and (7)).

The analysis type is linear static, as more refined analyses on one hand require additional data currently not available, and on the other hand they will not provide relevant benefits in terms of reducing the uncertainty of the final results. Four structural verifications are carried out:

1. Stresses verification: in the most stressed areas, to ensure that tensile and compressive stresses do not exceed material strength, appropriately reduced by a safety factor;
2. Section verification: at the inner base of the structure, considering only the axial force (N) and bending moment (M_x or M_y) at the most critical point;
3. Thrust line verification: in central sections of the structure, to confirm that the thrust line remains within the arch thickness, through Mery's method [115];
4. Verification under service condition: deflection analysis.

Stress check is done using a combined view, which represents invariant stresses independent of axis orientation. The analysis considers principal stresses 11 (maximum tension or minimum compression) and 22 (maximum compression or minimum tension) in shell elements. For the sake of clarity, the maximum stress values in both tension and compression were examined on the outer ($z+$) and inner ($z-$) surfaces of the habitat.

To account for uncertainties related to material performance and construction processes, in the final verification all computed stresses were amplified by a factor of 1.5, while the material strengths were reduced by a factor of 2. The 1.5 factor on stresses follows the classical load amplification approach defined in Eurocode EN 1990/EN 1992 [113], adopted to ensure the overall safety level of the structure. This factor accounts for possible load variations, geometric imperfections, modeling simplifications, and uncertainties in actual applied actions, ensuring that the design remains safe under conditions that may deviate from nominal assumptions. The factor of 2 on material strength was chosen conservatively. In the Eurocode [113], for standardized and well-characterized materials such as steel ($\gamma_m = 1.15$) or concrete ($\gamma_m = 1.5$), partial safety factors are rigorously defined based on statistical material properties and long-term empirical data. In contrast, our sulfurcrete represents an experimental material applied in an entirely unknown environment, with novel construction techniques and multiple sources of uncertainty, including variability of material properties, potential defects, and extreme environmental conditions. Therefore, the design strength was reduced by a factor of 2 to provide an additional safety margin, fully consistent with Eurocode principles and

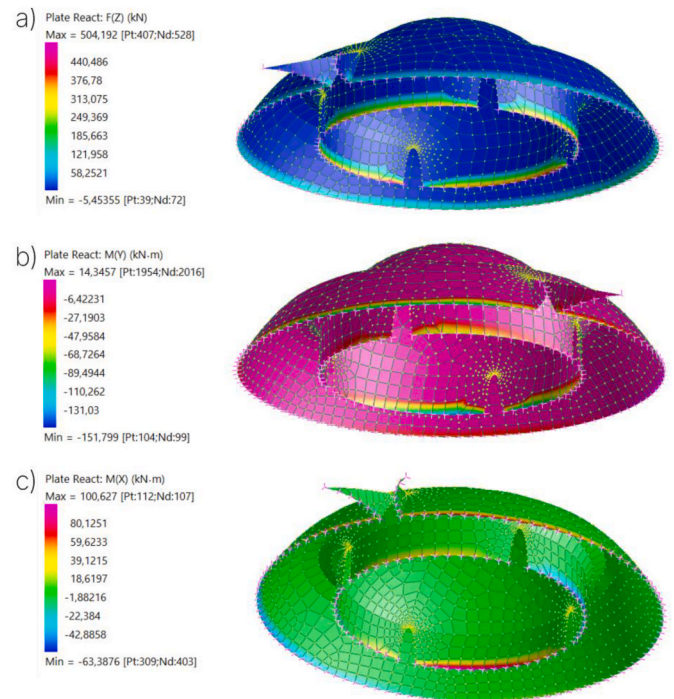


Fig. 18. a) Distribution of normal stresses along the surface of the mesh; b) distribution of bending moments in the y-direction over the mesh surface; c) distribution of bending moments in the x-direction over the mesh surface.

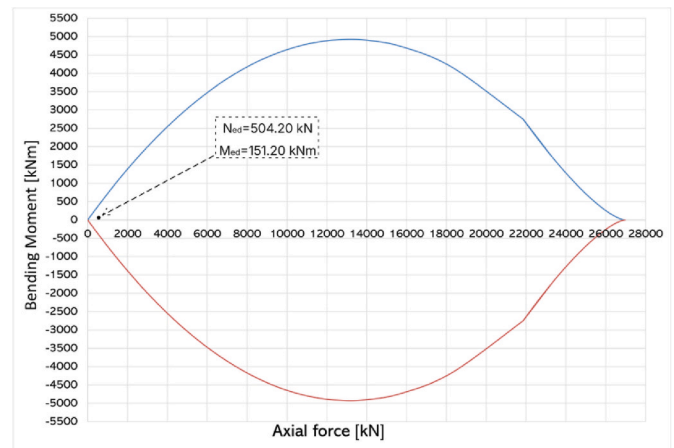


Fig. 19. N - M_y iteration domain for section and identification of the design stresses (the value represented by the black point corresponds to the observed value of designed load).

conservative structural design practices.

As Fig. 17 shows, the structure works predominantly in compression, consistent with the design concept and the mechanical properties of the material. The maximum compressive stress is experienced (0.696 MPa) near the arched opening to the inner dome. This value is extremely lower than the compressive strength of the material. For the parts of the structure subject to tension, the analysis of local vectors reveals a maximum stress of 0.231 MPa. Even in this case, the verification is satisfied with a high margin.

A combined bending-compression check was then performed at the most stressed regions of the structure. The verification is carried out comparing the demand and capacity in terms of axial force and bending moment, at the ground connection between the catenary torus and the central dome, see Fig. 18 representing the highly stressed region. The

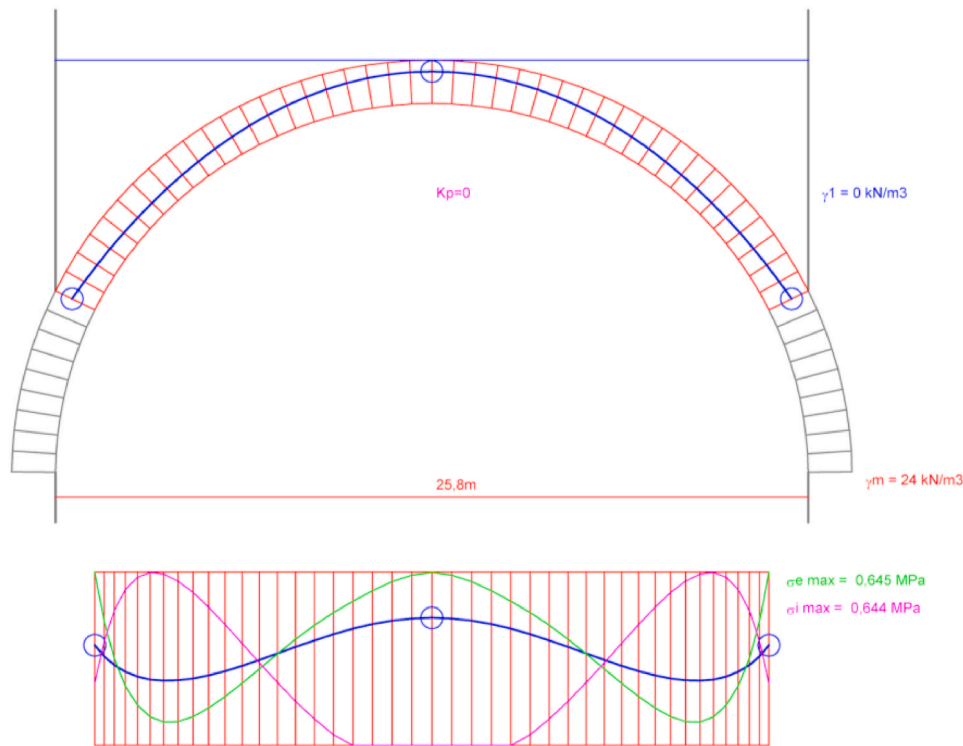


Fig. 20. Méry's funicular curve of our case study.

most stressed area of the roof exhibits an axial force $N = 504.19$ kN, bending moments $M_y = 151.80$ kN m and $M_x = 100.627$ kN m, that were verified adopting the interaction domain derived according to the material mechanical properties and section geometry. In detail, the section is considered with a rectangular section with base 1.5 m and height 1 m, and marscrete properties with a design compressive strength $f_{cd} = 19.5$ MPa (already considering partial safety factors). The section, under the design loads, is verified for combined bending and compression, and it demonstrates substantial structural reserves (see Fig. 19). Then, using Heyman's Plasticity Theorem [116], the position of the thrust line in arched sections of the structure, considering the self-weight load combination, was evaluated through a linear kinematic analysis. Particularly, this verification is done only to the central dome of the habitat, given its larger span and height compared to the surrounding semi-torus. Under the hypothesis that the external structure can be considered similar to a masonry vaulted structure made with rigid bodies subject to compression only (as demonstrated above), the following assumptions apply:

- The material does not resist to any tensile strength;
- The material has infinite compressive strength, and crushing phenomena are neglected;
- The material is infinitely rigid below the limit load and perfectly plastic at ultimate resistance;
- The collapse occurs through a kinematic mechanism, plastic hinges form at critical points.

It is recalled that the assumption of zero tensile strength is conservative, as marscrete exhibits a finite tensile strength value. The verification was performed using Méry's method (funicular curve), inputting geometric data and material density, setting any overlay layers to zero thickness and density, and modeling 60 ideal voussoirs. For completeness, a 1-m strip of the dome was considered with perfectly radial voussoirs, whereas the real membrane geometry and openings would influence behavior. This approximation is conservative, as it neglects interaction with the actual structural stiffness, providing additional

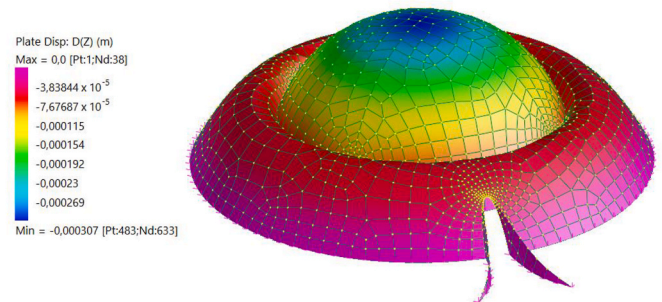


Fig. 21. Distribution of displacements in the global z-axis of the system.

safety margins. Results are shown in Fig. 20.

Finally, regarding displacement verification again evaluated considering the self-weight load only, they are negligible. Results show a maximum deflection of 0.3 mm at the dome edge (Fig. 21), over a span of approximately 25 m. This amount is considered negligible compared to conventional deformation limits (1/250 of the span or 1/1000 for strategic structures [112]). Although a full normative check is not feasible for such a unique structure, the results show substantial safety margins regarding the deformational performance.

4.5.2. Analysis of thermal stresses and heat flux gradients

To evaluate the structural behavior under large temperature fluctuations, a specific analysis has been carried out with a refined modeling strategy. The analysis has been carried out in the same finite element software Strand7 [111] used for the previous design. The structure was modeled using 1225 brick elements, extruded through the thickness from plate elements to ensure accurate results on both the upper (z+) and lower (z-) surfaces. The same material properties reported in Table 7, previously used for the linear analysis, were assigned, maintaining the same gravitational acceleration of 3.73 m/s^2 . The temperatures reported in Table 2 were imported as time-dependent factors and

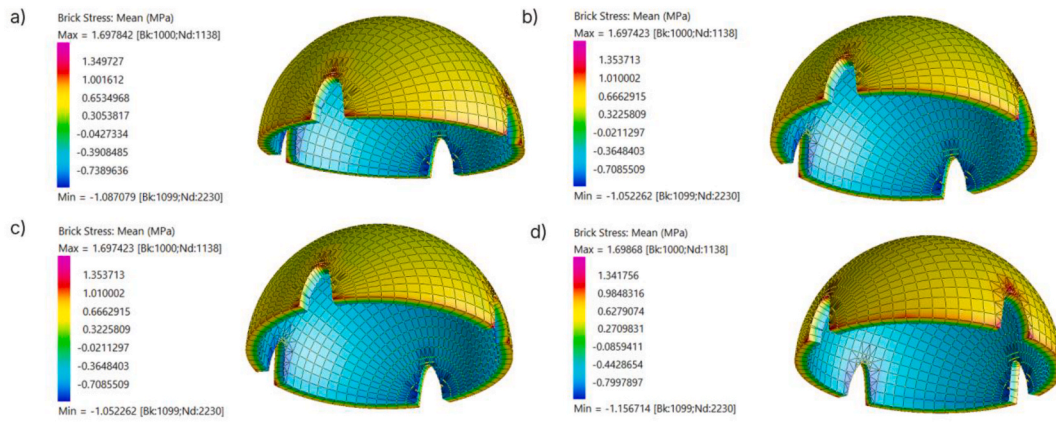


Fig. 22. Tensile stresses (positive values) and compressive stresses (negative values) occurring during the Martian equinoxes and solstices at 6 a.m.: a) Northern Hemisphere Spring Equinox b) Summer Solstice c) Winter Solstice d) Autumn Equinox.

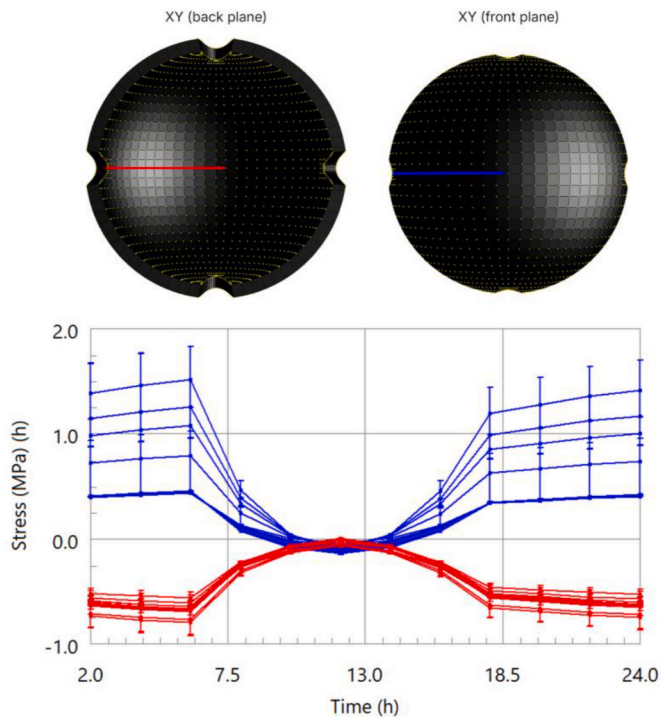


Fig. 23. Stress trends during the Autumn Equinox for the top fibers (blue) and bottom fibers (red). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

applied to the 1138 nodes on the external surface ($z+$). A constant temperature of 293.15 K was applied to the nodes on the internal surface of the dome, representing the ideal temperature condition to be maintained in the habitat. A quasi-static analysis is chosen to compute the variation of compressive and tensile stresses over the course of a day based on the temperature profile, allowing the identification of the worst-case scenario. It should be noted that the analysis is carried out as an approximation, exclusively on the central dome, since it represents the most relevant configuration for the case study, due to the large span of the symmetrically positioned openings. For completeness, these analyses should also be extended to include the semi-torus located at the perimeter of the circumference. As expected, the highest stress values -both tensile and compressive- occurred at 6 a.m. for all analyses performed. These values are shown in Fig. 22. For the case study under consideration, the Autumn Equinox represents the worst case, and stresses at the dome top fibers are almost entirely in tension, while the

Table 8

Maximum absolute tensile and compressive values obtained for the four analyzed cases.

	$L_s = 0^\circ$	$L_s = 90^\circ$	$L_s = 180^\circ$	$L_s = 270^\circ$
σ_C [MPa]	1.087	1.052	1.157	1.052
σ_T [MPa]	1.697	1.697	1.699	1.697

Table 9

Thermal parameters associated with the structure under investigation.

λ (Thermal Conductivity) [$\text{WK}^{-1}\text{m}^{-1}$] [117]	0.085
C_p (Volumetric Heat Capacity) [$\text{JK}^{-1}\text{m}^{-3}$] [117]	$1.05 \cdot 10^6$
Thickness [m]	1.5
α (Thermal Diffusivity) [m^2/s]	$8.10 \cdot 10^{-8}$
Period of External Temperature Variation (1 sol) [s]	88,775
Thermal Diffusion Characteristic Time (h)	2458.3
ω [rad/s]	$7.07 \cdot 10^{-5}$

bottom fibers of the dome remain entirely in a state of compressive stress, see Fig. 23. This insight provides a basis for further analyses under different thermal cycles, loading conditions, or alternative geometrical configurations. The stress verification once again proves to be positive, as the internal compressive and tensile stresses (Table 8) remain widely below the material's strength limits.

Thermal analysis is carried out not only for safety purposes, but also for investigating the insulant properties of the structure to guarantee comfort for the astronauts and energy consumption limitation. Using the same dataset shown in Tables 2, it is demonstrated that temperature variations occurring during solstices and equinoxes have virtually no effect on the internal temperature of the structure. All thermal parameters, dependent on the material properties and on the geometry of the structure, are summarized in Table 9. These parameters are necessary in order to determine the penetration depth (δ) of a thermal wave using equation Eq. (8), that is the characteristic distance over which a periodic thermal oscillation imposed at the surface propagates into a material before being attenuated to approximately $1/e$ of its initial amplitude. It arises from the solution of harmonic thermal diffusion and depends on the material's thermal diffusivity (α) and the frequency of the excitation (ω). Beyond 3–4 times this depth, the temperature variation becomes negligible. In essence, it quantifies the spatial scale over which a periodic thermal cycle can influence the interior of a solid body.

$$\delta = \sqrt{\frac{2\alpha}{\omega}} \quad (8)$$

The penetration depth of the thermal wave was found to be 4.78 cm. Since this value is roughly thirty times smaller than the total thickness of

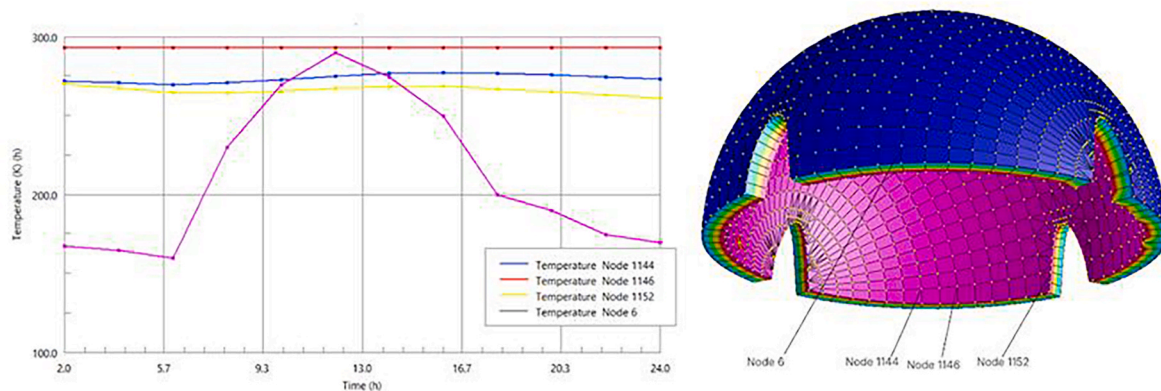


Fig. 24. Temperature trend during the Northern Hemisphere Spring Equinox for one external point (representative of the behavior of all external nodes) and three internal points within the structure (temperatures drop as we get closer to the openings). The nearly linear trend of the internal temperatures highlights that the external temperature does not influence the internal environment.

the structure, the interior is not affected by the daily external temperature fluctuations. Supporting the validity of these results, the finite element model confirms this behavior by showing that the internal points of the structure experience only negligible variations from the initial temperature, even when subjected to an external ΔT of 100 K (Fig. 24).

5. Conclusions

This paper has shown that a rigorous analysis of the Martian environment, combined with architectural and structural strategies, supports the feasibility of establishing a human outpost on the Red Planet. The design of habitable spaces, together with the use of innovative in-situ materials, suggests that life on Mars is technically conceivable, while acknowledging the uncertainties introduced by the sparse data on effects and load models, material and structural response under harsh environments. By defining this topic as a field that is still not fully developed, the main challenge encountered lay in the inherent arbitrariness -though reasoned and productive- of the mathematical and physical models used to describe the Martian environmental aspects. Added to this is the complexity of determining, using scientific criteria, the validity and acceptability of the observed structural behaviors. The concept underlying this work is not presented as a definitive solution, rather as an engineering framework that outlines distinct research domains – from the identification of a suitable landing site (Vernal Crater) to the development of radiation and micrometeoroid shielding systems, the validation of marscrete as a viable construction material, and the analysis of Martian seismic activity. The multidisciplinary nature of this paper has allowed us to propose the first ground motion attenuation law tailored to the Martian geological context. Hence, Mars emerges as both a realistic and necessary objective, continuing to drive scientific and technological advancement in the twenty-first century. While many critical aspects of an interplanetary mission remain unresolved, this work provides a structured, engineering-based methodology that can be further refined and expanded through future research.

Data availability statement

The data that support the findings of this study are available upon request from the authors.

CRediT authorship contribution statement

Pietro Lapo Moro: Data curation, Formal analysis, Investigation, Software, Visualization, Writing – original draft. **Klajdi Toska:** Software, Validation. **Mariano Angelo Zanini:** Data curation, Investigation, Methodology, Validation. **Maurizio Pajola:** Formal analysis,

Methodology, Supervision, Validation, Writing – review & editing. **Flora Faleschini:** Conceptualization, Methodology, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to thank Dr. Pierpaolo Ruttico from Politecnico di Milano, Italy, for the inspiring conversations and fruitful discussion on the application of robotics and additive manufacturing for extraterrestrial construction and explorations.

References

- [1] R. Schmidt, Mars express — ESA's first mission to planet Mars, *Acta Astronaut.* 52 (2–6) (2003) 197–202, [https://doi.org/10.1016/S0094-5765\(02\)00157-1](https://doi.org/10.1016/S0094-5765(02)00157-1).
- [2] R.W. Zurek, S.E. Smrekar, An overview of the mars reconnaissance orbiter (MRO) science mission, *J. Geophys. Res.* 112 (2007) E05S01, <https://doi.org/10.1029/2006JE002701>.
- [3] J. Vago, O. Witasse, H. Svedhem, P. Baglioni, A. Haldemann, G. Gianfiglio, T. Blancquaert, D. McCoy, R. de Groot, ESA ExoMars program: the next step in exploring Mars, *Sol. Syst. Res.* 49 (2015) 518–528, <https://doi.org/10.1134/S003809461507019>.
- [4] B.M. Jakosky, J.M. Grebowsky, J.G. Luhmann, D.A. Brain, Initial results from the MAVEN mission to Mars, *Geophys. Res. Lett.* 42 (21) (2015) 8791–8802, <https://doi.org/10.1002/2015GL065271>.
- [5] H.H. Kieffer, B.M. Jakosky, C.W. Snyder, M.S. Matthews (Eds.), *Mars*, University of Arizona Press, 1992, <https://doi.org/10.2307/j.ctt207g59v>.
- [6] European Space Agency, Mars' ionosphere shaped by crustal magnetic fields, <https://sci.esa.int/web/mars-express/-/58554-mars-ionosphere-shaped-by-crustal-magnetic-fields>, 2016, November 3.
- [7] R.P. Mueller, T.J. Prater, M. Roman, J.E. Edmunson, M.R. Fiske, P. Carrato, NASA Centennial Challenge: Three Dimensional (3D) Printed Habitat, Phase 3, 70th International Astronautical Congress (IAC), Washington, D.C., 2019. <https://it.scribd.com/document/670575692/nasa-centennial-challenge-three-dimensional-3d-printed-habitat-phase-3>.
- [8] T. Prater, N. Werkheiser, F. Ledbetter, D. Timucin, K. Wheeler, M. Snyder, 3D printing in zero G technology demonstration mission: complete experimental results and summary of related material modeling efforts, *Int. J. Adv. Manuf. Technol.* 101 (1–4) (2019) 391–417, <https://doi.org/10.1007/s00170-018-2827-7>.
- [9] C. Clinton, T. Prater, N. Werkheiser, K. Morgan, F. Ledbetter, NASA additive manufacturing initiatives for deep space exploration, in: *Proceedings of the 69th International Astronautical Congress (IAC)*, Bremen, Germany, 2018, October 1–5.
- [10] O.K. Soureshjani, A. Massumi, Martian buildings: structural forms using in-place sources, *Sci. Rep.* 12 (2022) 21992, <https://doi.org/10.1038/s41598-022-25507-5>.

- [11] C. Zhou, S. Cheng, Y. Gao, J. Zhao, L. Xiao, L. Ding, Zhuque base for Martian habitation: conceptual design and performance analysis of cave dwellings and in situ construction, *Research* 8 (2025), <https://doi.org/10.34133/research.0849>, Article 0849.
- [12] E. Koca, A. Turer, Design, optimization, and autonomous construction of Martian infrastructure including slabs using in-situ CO₂-based polyethylene, *J. Build. Eng.* 98 (2024) 111318, <https://doi.org/10.1016/j.jobe.2024.111318>.
- [13] G. Etiopie, D.Z. Oehler, Methane spikes, background seasonality and non-detections on Mars: a geological perspective, *Planet. Space Sci.* 168 (2019) 52–61, <https://doi.org/10.1016/j.pss.2019.02.001>.
- [14] M.H. Carr, G.D. Clow, Martian channels and valleys: their characteristics, distribution, and age, *Icarus* 48 (1) (1981) 91–117, [https://doi.org/10.1016/0019-1035\(81\)90156-1](https://doi.org/10.1016/0019-1035(81)90156-1).
- [15] J.N. James, The voyage of mariner IV, *Sci. Am.* 214 (3) (1966) 42–53, <http://www.jstor.org/stable/24931296>.
- [16] G.A. Soffen, The viking project, *J. Geophys. Res.* 82 (28) (1977) 3959–3970, <https://doi.org/10.1029/J082i028p03959>.
- [17] R. Sagdeev, A. Zakharov, Brief history of the phobos mission, *Nature* 341 (1989) 581–585, <https://doi.org/10.1038/341581a0>.
- [18] D.T. Lyons, J.G. Beerer, P. Esposito, M.D. Johnston, W.H. Willcockson, Mars global surveyor: aerobraking mission overview, *J. Spacecraft Rockets* 36 (3) (1999) 307–315, <https://doi.org/10.2514/2.3472>.
- [19] R.S. Saunders, R.E. Arvidson, G.D. Badhwar, W.V. Boynton, P.R. Christensen, F. A. Cucinotta, W.C. Feldman, R.G. Gibbs, Jr. C. Kloss, M.R. Landano, R.A. Mase, G. W. McSmith, M.A. Meyer, I.G. Mitrofanov, G.D. Pace, J.J. Plaut, W.P. Sidney, D. A. Spencer, T.W. Thompson, C.J. Zeitlin, 2001 Mars odyssey mission summary, *Space Sci. Rev.* 110 (1–2) (2004) 1–36, <https://doi.org/10.1023/B:SPAC.0000021006.84299.18>.
- [20] L.A. D'Amario, Mars exploration rovers navigation results, *J. Astronaut. Sci.* 54 (2006) 129–173, <https://doi.org/10.1007/BF03256481>.
- [21] P.H. Smith, The phoenix mission to Mars, in: 2004 IEEE Aerospace Conference Proceedings, vol. 1, 2004, pp. 337–341, <https://doi.org/10.1109/AERO.2004.1367617>.
- [22] A.R. Vasavada, Mission overview and scientific contributions from the Mars science laboratory curiosity rover after eight years of surface operations, *Space Sci. Rev.* 218 (2022) 14, <https://doi.org/10.1007/s11214-022-00882-7>.
- [23] W. Xu, X. Liu, Z. Yan, et al., The MarSCoDe instrument suite on the Mars rover of china's Tianwen-1 mission, *Space Sci. Rev.* 217 (2021) 64, <https://doi.org/10.1007/s11214-021-00836-5>.
- [24] H. Almatroushi, H. AlMazmi, N. AlMheiri, et al., Emirates Mars mission characterization of Mars atmosphere dynamics and processes, *Space Sci. Rev.* 217 (2021) 89, <https://doi.org/10.1007/s11214-021-00851-6>.
- [25] The Planetary Society. (n.d.). Every mission to Mars ever. <https://www.planetary.org/space-missions/every-mars-mission>.
- [26] C. Leovy, Weather and climate on Mars, *Nature* 412 (2001) 245–249, <https://doi.org/10.1038/35084192>.
- [27] F. Hourdin, P. Le Van, F. Forget, O. Talagrand, Meteorological variability and the annual surface pressure cycle on Mars, *J. Atmos. Sci.* 50 (21) (1993) 3625–3640, [https://doi.org/10.1175/1520-0469\(1993\)050<3625:MVTAS>2.0.CO;2](https://doi.org/10.1175/1520-0469(1993)050<3625:MVTAS>2.0.CO;2).
- [28] B. Knapmeyer-Endrun, M.P. Panning, F. Bissig, R. Joshi, A. Khan, V. Lekic, W. B. Banerdt, Thickness and structure of the Martian crust from InSight seismic data, *Science* 373 (6553) (2021) 438–443, <https://doi.org/10.1126/science.abf8966>.
- [29] G. Leone, A network of lava tubes as the origin of Labyrinthus noctis and valles marineris on Mars, *J. Volcanol. Geoth. Res.* 277 (2014) 1–8, <https://doi.org/10.1016/j.jvolgeores.2014.01.011>.
- [30] B.G. Drake, W.D. Watts, Human exploration of Mars, design reference architecture 5.0, Addendum #2. NASA Technical Reports Server (NTRS) (2014). <https://ntrs.nasa.gov/citations/20160003093>.
- [31] M. Golombek, J. Grant, D. Kipp, et al., Selection of the Mars science laboratory landing site, *Space Sci. Rev.* 170 (2012) 641–737, <https://doi.org/10.1007/s11214-012-9916-y>.
- [32] M. Pajola, S. Rossato, E. Baratti, R. Pozzobon, C. Quantin, J. Carter, P. Thollot, *The simud-tiu valles hydrologic system: a multidisciplinary study of a possible site for future Mars on-site exploration*, *Icarus* 268 (2016) 355–381, <https://doi.org/10.1016/j.icarus.2015.12.049>.
- [33] M. Pajola, S. Rossato, J. Carter, E. Baratti, R. Pozzobon, M.S. Erculiani, M. Coradini, K. McBride, Eridania basin: an ancient paleolake floor as the next landing site for the Mars 2020 rover, *Icarus* 275 (2016) 163–182, <https://doi.org/10.1016/j.icarus.2016.03.029>.
- [34] M. Pajola, S. Rossato, E. Baratti, A. Kling, Planetary mapping for landing sites selection: the Mars case study, in: *Planetary Cartography and GIS. Lecture Notes in Geoinformation and Cartography*, Springer, Cham, 2019, https://doi.org/10.1007/978-3-319-62849-3_7.
- [35] V. Schorbach, T. Weiland, Wind as a back-up energy source for Mars missions, *Acta Astronaut.* 191 (2022) 472–478, <https://doi.org/10.1016/j.actaastro.2021.11.013>.
- [36] B. Jackson, L. Fenton, T. Brown, A. Munguira, G. Martinez, C. Newman, D. Viúdez-Moreiras, M. Golombek, R. Lorenz, M.D. Paton, D. Conway, Profiling near-surface winds on Mars using attitude data from Mars 2020 ingenuity, *Planet. Sci. J.* 6 (1) (2025) 21, <https://doi.org/10.3847/psj/ad8b41>.
- [37] T.B. Kimberlain, E.S. Blake, J.P. Cangialosi, Tropical Cyclone Report: Hurricane Patricia (20–24 October 2015), National Hurricane Center, 2016.
- [38] K.B. Hille, The Fact and Fiction of Martian Dust Storms, NASA, 2023. <https://www.nasa.gov/solar-system/the-fact-and-fiction-of-martian-dust-storms/#:~:text=The%20winds%20in%20the%20strongest,as%20dense%20as%20Earth's%20atmosphere>.
- [39] I.J. Daubar, C.M. Dundas, A.S. McEwen, A. Gao, D. Wexler, S. Piqueux, et al., New craters on Mars: an updated catalog, *J. Geophys. Res. Planets* 127 (2022), <https://doi.org/10.1029/2021JE007145>.
- [40] A. Borgegräfe, M. Quatmann, D. Nölke, Radiation protective structures on the base of a case study for a manned Mars mission, *Acta Astronaut.* 65 (9–10) (2009) 1292–1305, <https://doi.org/10.1016/j.actaastro.2009.03.025>.
- [41] M.H. Carr, J.W. Head, Geologic history of Mars, *Earth Planet. Sci. Lett.* 294 (3–4) (2010) 185–203, <https://doi.org/10.1016/j.epsl.2009.06.042>.
- [42] A. Yin, Structural analysis of the valles marineris fault zone: possible evidence for large-scale strike-slip faulting on Mars, *Lithosphere* 4 (4) (2012) 286–330, <https://doi.org/10.1130/L192.1>.
- [43] Iris DMC. (n.d.). MARS events: insight SEIS seismic event catalog. Retrieved from <http://ds.iris.edu/ds/nodes/dmc/tools/mars-events/>.
- [44] X. Wang, L. Chen, X. Wang, Renewed epicentral distribution of low-frequency marsquakes by varying-parameter polarization analysis of InSight data, *Geophys. Res. Lett.* 50 (2023), <https://doi.org/10.1029/2023GL103896>.
- [45] D.G. Horvath, P. Moitra, C.W. Hamilton, R.A. Craddock, J.C. Andrews-Hanna, Evidence for geologically recent explosive volcanism in elysium planitia, Mars, *Icarus* 365 (2021) 114499, <https://doi.org/10.1016/j.icarus.2021.114499>.
- [46] J.A. Taylor, N.A. Teanby, J. Wookey, Estimates of seismic activity in the Cerberus fossae region of Mars, *J. Geophys. Res. Planets* 118 (2) (2013), <https://doi.org/10.1002/2013JE004469>.
- [47] C. Perrin, M. Plasman, N.A. Teanby, S.E. Smrekar, D. Giardini, Geometry and segmentation of Cerberus fossae, Mars: implications for marsquake properties, *J. Geophys. Res. Planets* 127 (11) (2022), <https://doi.org/10.1029/2021JE007118>.
- [48] J. Vetterlein, G.P. Roberts, Structural evolution of the northern Cerberus fossae graben system, elysium planitia, Mars, *J. Struct. Geol.* 32 (4) (2010) 394–406, <https://doi.org/10.1016/j.jsg.2009.11.004>.
- [49] S.C. Stähler, A. Khan, W.B. Banerdt, P. Lognonné, D. Giardini, S. Ceylan, M. Drilleau, R.F. Garcia, A.C. Duran, Q. Huang, D. Kim, V. Lekic, H. Samuel, N. Schmerr, D. Sollberger, E. Stutzmann, Z. Xu, C. Charalambous, P.M. Davis, D. Antonangeli, J.C.E. Irving, M. Knapmeyer, T. Kawamura, R. Maguire, A. Marusiak, M. Panning, C. Perrin, A.-C. Plesa, A. Rivoldini, C. Schmelzbach, G. Zenhäusern, É. Beucler, J. Clinton, N.L. Dahmen, M. van Driel, T. Gudkova, A. Horleston, T. Pike, M. Plasman, S. Smrekar, Seismic detection of the Martian core, *Science* 373 (6553) (2021) 443–448, <https://doi.org/10.1126/science.abi7730>.
- [50] M. Pajola, R. Pozzobon, S. Silvestro, F. Salese, S. Rossato, L. Pompilio, G. Munaretto, L. Teodoro, A. Kling, E. Simioni, A. Lucchetti, L. Tornabene, L. Marinangeli, A. Tangari, J. Wilson, G. Cremonese, M. Massironi, N. Thomas, Geology, in-situ resource-identification and engineering analysis of the vernal crater area (Arabia terra): a suitable Mars human landing site candidate, *Planet. Space Sci.* 213 (2022) 105444, <https://doi.org/10.1016/j.pss.2022.105444>.
- [51] G. Munaretto, M. Pajola, G. Cremonese, C. Re, A. Lucchetti, E. Simioni, A. S. McEwen, A. Pommerol, P. Becerra, S.J. Conway, N. Thomas, M. Massironi, Implications for the origin and evolution of Martian recurring slope lineae at hale crater from CaSSIS observations, *Planet. Space Sci.* 187 (2020) 104947, <https://doi.org/10.1016/j.pss.2020.104947>.
- [52] L. Wan, R. Wendner, G. Cusatis, A novel material for in situ construction on Mars: experiments and numerical simulations, *Constr. Build. Mater.* 120 (2016) 222–231, <https://doi.org/10.1016/j.conbuildmat.2016.05.046>.
- [53] I. Giwa, M. Dempsey, M. Fiske, A. Kazemian, 3D printed sulfur-regolith concrete performance evaluation for waterless extraterrestrial robotic construction, *Autom. Constr.* 165 (2024) 105571, <https://doi.org/10.1016/j.autcon.2024.105571>.
- [54] Z. Hu, T. Shi, M. Cen, J. Wang, X. Zhao, C. Zeng, Y. Zhou, Y. Fan, Y. Liu, Z. Zhao, et al., Research progress on lunar and Martian concrete, *Constr. Build. Mater.* 343 (2022) 128117, <https://doi.org/10.1016/j.conbuildmat.2022.128117>.
- [55] M.H. Shahsavari, M.M. Karbala, S. Iranfar, V. Vandeginste, Martian and lunar sulfur concrete mechanical and chemical properties considering regolith ingredients and sublimation, *Constr. Build. Mater.* 350 (2022) 128914, <https://doi.org/10.1016/j.conbuildmat.2022.128914>.
- [56] J.M. Dale, A.C. Ludwig, Feasibility Study for Using sulfur-aggregate Mixtures as Structural Material (Technical Report No. AFWL-TR-66-57), Southwest Research Institute, 1966.
- [57] H.A. Omar, Production of Lunar Concrete Using Molten Sulfur (Final Research Report for JoVe NASA Grant NAG8-278), University of South Alabama, 1993.
- [58] Y. Reches, Concrete on Mars: options, challenges, and solutions for binder-based construction on the red planet, *Cement Concr. Compos.* 104 (2019) 103349, <https://doi.org/10.1016/j.cemconcomp.2019.103349>.
- [59] R. Boyd, P. Thompson, B. Clark, Duricrete and composites construction on mars, The case for mars iii: Strategies for exploration- general interest and overview (a 90-16651 05-12). San diego, CA, Univelt, inc., 1989 74 (1989).
- [60] J. Gifford, M.M. Hassan, T. Rupnow, M. Barbato, A. Okeil, S. Asadi, Dicyclopentadiene and sodium silicate microencapsulation for self-healing of concrete, *J. Mater. Civ. Eng.* 26 (5) (2014) 886–896, [https://doi.org/10.1061/\(ASCE\)MT.1943-5533.0000892](https://doi.org/10.1061/(ASCE)MT.1943-5533.0000892).
- [61] M. Gernhardt, S. Chappell, K. Beaton, H. Litaker, O. Bekdash, C. Newton, J. Stoffel, Deep Space Habitability Design Guidelines Based on the NASA NextSTEP Phase 2 Ground Test Program (NASA/TP-2020-220505), NASA Johnson Space Center, 2019.
- [62] Nasa, CHAPEA, Nasa (2025). <https://www.nasa.gov/humans-in-space/chaepa/>.

- [63] D.Z. Oehler, C.C. Allen, Vernal crater, SW arabia terra: MSL candidate with extensively layered sediments, possible Lake deposits, and a long history of subsurface ice, NASA Technical Reports Server (NTRS) (2007). <https://ntrs.nasa.gov/citations/20070017885>.
- [64] E.P. Steinberg, W. Bulleit, Reliability analysis of meteoroid loading on lunar structures, *Struct. Saf.* 15 (1–2) (1994) 51–66.
- [65] C. Patiño, S. Ruiz, D. Gomez, A. Cruz, S.J. Dyke, J. Ramirez, Seismic assessment of a long-term lunar habitat, *Acta Astronaut.* 214 (2024) 414–426.
- [66] NASA, NASA Space Flight human-system Standard: Spacecraft Maximum Allowable Concentrations of Airborne Contaminants (NASA STD 3001), NASA, 2019.
- [67] H.J. Llamas, K.L. Aplin, L. Berthoud, Effectiveness of Martian regolith as a radiation shield, *Planet. Space Sci.* 218 (2022) 105517, <https://doi.org/10.1016/j.pss.2022.105517>.
- [68] G. Genta, Next Stop Mars, Springer, Chichester, United Kingdom, 2017.
- [69] B. Khoshnevis, A. Carlson, M. Thangavelu, ISRU-Based Robotic Construction Technologies for Lunar and Martian Infrastructures (Report No. HQ-E-DAATN41353), NASA, 2017.
- [70] M.I. Dobynde, Y.Y. Shprits, A.Y. Drozdov, J. Hoffman, J. Li, Beating 1 sievert: optimal radiation shielding of astronauts on a mission to Mars, *Space Weather* 19 (2021), <https://doi.org/10.1029/2021SW002749> e2021SW002749.
- [71] M. Durante, Space radiation protection: destination Mars, *Life Sci. Space Res.* 1 (2014) 2–9, <https://doi.org/10.1016/j.lssr.2014.01.002>.
- [72] T.C. Slaba, C.J. Mertens, S.R. Blattning, Radiation shielding optimization on Mars, NASA Technical Reports Server (NTRS) (2013, April 1). <https://ntrs.nasa.gov/citations/20130012456>.
- [73] D. Gakis, D. Atri, Modeling the effectiveness of radiation shielding materials for astronaut protection on Mars, *The European Physical Journal Plus* 139 (8) (2024), <https://doi.org/10.1140/epjp/s13360-024-05519-y>.
- [74] S. Ceylan, J.F. Clinton, D. Giardini, S.C. Stähler, A. Horleston, T. Kawamura, M. Böse, C. Charalambous, N.L. Dahmen, M. Van Driel, C. Durán, F. Euchner, A. Khan, D. Kim, M. Plasman, J. Scholz, G. Zenhäusern, E. Beucier, R.F. Garcia, et al., The marsquake catalogue from InSight, sols 0–1011, *Phys. Earth Planet. Inter.* 333 (2022) 106943, <https://doi.org/10.1016/j.pepi.2022.106943>.
- [75] N.L. Dahmen, J.F. Clinton, M.A. Meier, S.C. Stähler, S. Ceylan, D. Kim, et al., MarsQuakeNet: a more complete marsquake catalog obtained by deep learning techniques, *J. Geophys. Res. Planets* 127 (2022), <https://doi.org/10.1029/2022JE007503>.
- [76] M.A. Zanini, L. Hofer, F. Faleschini, Reversible ground motion-to-intensity conversion equations based on the EMS-98 scale, *Eng. Struct.* 180 (2019) 310–320, <https://doi.org/10.1016/j.engstruct.2018.11.032>.
- [77] M. Drilleau, H. Samuel, R.F. Garcia, A. Rivoldini, C. Perrin, C. Michaut, et al., Marsquake locations and 1-D seismic models for Mars from InSight data, *J. Geophys. Res. Planets* 127 (2022), <https://doi.org/10.1029/2021JE007067> e2021JE007067.
- [78] N. Brinkman, J.F. Clinton, D. Giardini, P. Lognonné, S.C. Stähler, T. Kawamura, W.B. Banerdt, First focal mechanisms of marsquakes, *J. Geophys. Res. Planets* 126 (3) (2021), <https://doi.org/10.1029/2020JE006546> e2020JE006546.
- [79] C. Durán, A. Khan, S. Ceylan, G. Zenhäusern, S. Stähler, J.F. Clinton, D. Giardini, Seismology on Mars: an analysis of direct, reflected, and converted seismic body waves with implications for interior structure, *Phys. Earth Planet. Inter.* 325 (2022) 106851, <https://doi.org/10.1016/j.pepi.2022.106851>.
- [80] R.M.W. Musson, P.C. Marrow, P.W. Winter, Attenuation of Earthquake Ground Motion in the UK. Technical Report AEA/CS/16422000/ZJ745/004, AEA Technology Consultancy Services (SRD) and British Geological Survey, 1994, May.
- [81] C. Radu, D. Lungu, S. Demetriu, O. Coman, Recurrence, attenuation and dynamic amplification for intermediate depth Vrancea earthquakes, *Proceedings of the XXIV General Assembly of the ESC III* (1994, Sep) 1736–1745.
- [82] D. Lungu, O. Coman, T. Moldoveanu, Hazard analysis for Vrancea earthquakes. Application to cernavoda NPP site in Romania, in: *Proceedings of the 13th International Conference on Structural Mechanics in Reactor Technology*, Division K, 1995, Aug, p. 538.
- [83] D. Lungu, T. Cornea, I. Craileanu, S. Demetriu, Probabilistic seismic hazard analysis for inelastic structures on soft soils, in: *In Proceedings of Eleventh World Conference on Earthquake Engineering*, 1996, Jun.
- [84] G.L. Molas, F. Yamazaki, Attenuation of earthquake ground motion in Japan including deep focus events, *Bull. Seismol. Soc. Am.* 85 (5) (1995) 1343–1358, <https://doi.org/10.1785/BSSA0850051343>.
- [85] R.P. Singh, A. Aman, Y.J.J. Prasad, Attenuation relations for strong seismic ground motion in the himalayan region, *Pure Appl. Geophys.* 147 (1) (1996) 161–180, <https://doi.org/10.1007/BF00876321>.
- [86] C.-E. Baag, S.-J. Chang, N.-D. Jo, J.-S. Shin, Evaluation of seismic hazard in the southern part of Korea, in: *Proceedings of the Second International Symposium on Seismic Hazards and Ground Motion in the Region of Moderate Seismicity*, 1998.
- [87] C. Reyes, *El Estado Limite De Servicio En El Diseño Sísmico De Edificios* (Ph.D. Thesis, School of Engineering, National Autonomous University of Mexico (UNAM), 1998 in Spanish).
- [88] M.C. Chapman, On the use of elastic input energy for seismic hazard analysis, *Earthq. Spectra* 15 (4) (1999) 607–635, <https://doi.org/10.1193/1.1586105>.
- [89] B.-Q. Wang, F.T. Wu, Y.J. Bian, Attenuation characteristics of peak acceleration in North China and comparison with those in the eastern part of North America, *Acta Seismol. Sin.* 12 (1) (1999) 26–34, <https://doi.org/10.1111/j.1365-246X.1999.tb01515>.
- [90] J. Xiang, D. Gao, The attenuation law of horizontal peak acceleration on the rock site in Yunnan area, *Earthq. Res. China* 8 (4) (1994) 509–516.
- [91] N.P. Theodulidis, B.C. Papazachos, Dependence of strong ground motion on magnitude-distance, site geology and macroseismic intensity for shallow earthquakes in Greece: I. Peak horizontal acceleration, velocity and displacement, *Soil Dynam. Earthq. Eng.* 11 (1992) 387–402, [https://doi.org/10.1016/02677261\(92\)90032-9](https://doi.org/10.1016/02677261(92)90032-9).
- [92] M. Garcia-Fernandez, J.A. Canas, Estimation of regional values of peak ground acceleration from short-period seismograms, *Proceedings of the Fourth International Conference on Seismic Zonation II* (1991) 533–539.
- [93] M. Garcia-Fernandez, J.A. Canas, Regional peak ground acceleration estimates in the Iberian peninsula, *Proceedings of the Fifth International Conference on Seismic Zonation II* (1995) 1029–1034.
- [94] M.K. Sen, Deep structural complexity and site response in Los Angeles basin, *Proceedings of the Fourth U.S. National Conference on Earthquake Engineering I* (1990) 545–553.
- [95] Y. Fukushima, T. Tanaka, S. Kataoka, A new attenuation relationship for peak ground acceleration derived from strong-motion accelerograms, *Proceedings of Ninth World Conference on Earthquake Engineering II* (1988) 343–348.
- [96] Y. Fukushima, T. Tanaka, A new attenuation relation for peak horizontal acceleration of strong earthquake ground motion in Japan, *Bull. Seismol. Soc. Am.* 80 (4) (1990) 757–783, <https://doi.org/10.1785/BSSA0800040757>.
- [97] S.K. Singh, E. Mena, R. Castro, C. Carmona, Empirical prediction of ground motion in Mexico City from coastal earthquakes, *Bull. Seismol. Soc. Am.* 77 (5) (1987) 1862–1867, <https://doi.org/10.1785/BSSA0770051862>.
- [98] E.L. Krinitzky, F.K. Chang, O.W. Nuttli, State-of-the-art for assessing earthquake hazards in the United States; report 26, parameters for specifying magnitude-related earthquake ground motions, Technical Report U.S. Army Engineer Waterways Experimental Station, Miscellaneous Paper S-73-1 (1987, Sep).
- [99] E.L. Krinitzky, F.K. Chang, O.W. Nuttli, Magnitude-related earthquake ground motions, *Bull. Assoc. Eng. Geol.* 25 (4) (1988) 399–423.
- [100] Advanced Textiles Association (ATA), Inflatable Space Habitats Undergo Testing, *Fabric Architecture Magazine*, 2023, November 1. <https://fabricarchitecturemag.com/2023/11/01/inflatable-space-habitats-undergo-testing/>.
- [101] G.D. Valle, D.A. Litteken, T.C. Jones, Review of habitable softgoods inflatable design, analysis, testing, and potential space applications, in: *Paper Presented at the AIAA Scitech Forum*, San Diego, CA, 2019, <https://doi.org/10.2514/6.2019-1018>.
- [102] J.M.C. Puguán, D.B.A. Agbayani, A.J.V. Sadural, H. Kim, Enhanced low frequency sound absorption of bilayer nanocomposite acoustic absorber laminated with microperforated nanofiber felt, *ACS Appl. Polym. Mater.* 5 (11) (2023) 8880–8889, <https://doi.org/10.1021/acsapm.3c01249>.
- [103] R.L. Kobrick, J.H. Agui, Preparing for planetary surface exploration by measuring habitat dust intrusion with filter tests during an analogue Mars mission, *Acta Astronaut.* 160 (2019) 297–309, <https://doi.org/10.1016/j.actaastro.2019.04.040>.
- [104] C. Wajochen, M. Münch, S. Kobialka, K. Kräuchi, R. Steiner, P. Oelhafen, S. Orgül, A. Cirjz-Justice, High sensitivity of human melatonin, alertness, thermoregulation, and heart rate to short wavelength light, *The Journal of Clinical Endocrinology & Metabolism* 90 (3) (2005) 1311–1316, <https://doi.org/10.1210/jc.2004-0957>.
- [105] A.U. Viola, L.M. James, L.J.M. Schlangen, D.J. Dijk, Blue-enriched white light in the workplace improves self-reported alertness, performance and sleep quality, *Scand. J. Work. Environ. Health* 34 (4) (2008) 297–306, <https://doi.org/10.5271/sjweh.1268>.
- [106] J.L. Wang, J.J. Rosenbaum, A.N. Prasad, R.R. Raad, E.J. Putman, A.D. Harrington, et al., Potential health impacts, treatments, and countermeasures of Martian dust on future human space exploration, *GeoHealth* 9 (2025), <https://doi.org/10.1029/2024GH001213> e2024GH001213.
- [107] ISO 14644 [ISO, ISO 14644-1: Cleanrooms and Associated Controlled Environments – Part 1: Classification of Air Cleanliness by Particle Concentration, International Organization for Standardization, 2015. <https://www.iso.org/standard/53394.html>.
- [108] M. Shahrjerdi, AI-driven cleanroom design for Mars: revolutionizing interplanetary habitats for 2050 colonies, *World J. Adv. Res. Rev.* 27 (3) (2025) 563–568, <https://doi.org/10.30574/wjarr.2025.27.3.3171>.
- [109] D. Vrankar, C. Verseux, C. Heinicke, An airlock concept to reduce contamination risks during the human exploration of Mars, *npj Microgravity* 9 (81) (2023). <https://www.nature.com/articles/s41526-023-00329-5>.
- [110] H.W. Jones, E.W. Hodgson, M.H. Kliss, *Life support for deep space and Mars* (ICES-2014-074), 44th international conference on environmental systems, NASA Technical Reports Server (2014, July). <https://ntrs.nasa.gov/api/citations/2014001164/downloads/2014001164.pdf>.
- [111] *Strand7* (version 3.1.1) [software]. Strand7 Pty Ltd. <https://www.strand7.com>.
- [112] Ministero delle Infrastrutture e dei Trasporti (MIT), Norme Tecniche per Le Costruzioni 2018 (Decreto Ministeriale 17 Gennaio 2018), *Gazzetta Ufficiale della Repubblica*, Italiana, 2018.
- [113] CEN - European Committee for Standardization, *Eurocode 0: Basis of Structural Design* (EN 1990), CEN, 2002.
- [114] CEN - European Committee for Standardization, *Eurocode 8: Design of Structures for Earthquake Resistance* (EN 1998), CEN, 2004.
- [115] E. Méry, Sur l'équilibre des voûtes en berceau, *Ann. Ponts Chaussées* 19 (1840) 50–70. I Sem.
- [116] J. Heyman, *The Stone Skeleton: Structural Engineering of Masonry Architecture*, Cambridge University Press, 1995.
- [117] E.S. Hütter, *Development and Testing of Thermal Sensors for Planetary Applications* (Doctoral Dissertation, karl-franzens-universität Graz), 2011.